Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Sea Level TAC

MDT products:

SEALEVEL_GLO_PHY_MDT_008_063

SEALEVEL_MED_PHY_MDT_L4_STATIC_008_066

SEALEVEL_BLK_PHY_MDT_L4_STATIC_008_067

SEALEVEL_EUR_PHY_MDT_L4_STATIC_008_070

Issue: 4.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	15/01/2021	All	Creation	S. Mulet; S. Jousset	M.I. Pujol
2.0	10/09/2021		Updated SEALEVEL_GLO_PHY_MDT_008_063 product	S. Jousset	M.I. Pujol
3.0	07/07/2023		Updated SEALEVEL_GLO_PHY_MDT_008_063 product	S. Jousset	M.I. Pujol
4.0	31/05/2024	I II IV.4	Added SEALEVEL_EUR_PHY_MDT_L4_STATIC_008_070 product (update of section I.1.2 and add section V.4)	S. Jousset	M.I. Pujol
5.0	09/10/2024	All	Major revision of the document to adapt it to the QUID 1.2 template.	S. Jousset	M.I Pujol

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I EXECUTIVE SUMMARY

This document presents the data processing and the qualification of the MDT product(s) listed bellow.

I.1 Products covered by this document

I.1.1 Global Product

The main caractheristics of the global product are described in Table 1.

Table 1: Global MDT characteristics.

Product	SEALEVEL_GLO_PHY_MDT_008_063 (Global MDT_HYBRID_CNES_CLS22_CMEMS2020)
Area	Global
Satellites	-
Spatial resolution	0.125° x 0.125°
Temporal resolution	20-year mean reference

Note about the covered area:

- Everywhere except in the Mediterranean Sea and in the Black Sea, the Mean Dynamic Topography (MDT) is the MDT_CNES_CLS22.
  - The MDT in the Mediterranean Sea comes from MDT_CMEMS_2020_MED, described in §I.1.2 and interpolated on 1/8° global MDT grid.
  - The MDT in the Black Sea comes from the MDT_CMEMS_2020_BLK, described in §I.1.2 and interpolated on 1/8° global MDT grid.

- The Arctic is processed with the global product, but its data processing is a little different and will be explained in paragraph III.2.

I.1.2 Regional products

The main caractheristics of the regional product are described in Table 2, Table 3 and Table 4.

Table 2: Mediterranean MDT characteristics.

Product	SEALEVEL_MED_PHY_MDT_L4_STATIC_008_066 (MDT_CMEMS_2020_MED)
Area	Mediterranean Sea
Satellites	-
Spatial resolution	0.04166° x 0.04166°
Temporal resolution	20-year mean reference

Table 3: Black Sea MDT characteristics.

Product	SEALEVEL_BLK_PHY_MDT_L4_STATIC_008_067 (MDT_CMEMS_2020_BLK)
Area	Black Sea (no information on Azov Sea and Marmara Sea)
Satellites	-
Spatial resolution	0.0625° x 0.0625°
Temporal resolution	20-year mean reference

Table 4: Europe MDT characteristics.

Product	SEALEVEL_EUR_PHY_MDT_L4_STATIC_008_070 (MDT_CMEMS_2024_EUR)
Area	European coasts
Satellites	-
Spatial resolution	0.0625° x 0.0625°
Temporal resolution	20-year mean reference

I.2 Summary of the results

Global MDT_HYBRID_CNES_CLS22_CMEMS2020

The main improvement of this new CNES-CLS22 MDT over the previous CNES-CLS18 MDT is in the Arctic, with better coverage, no artifacts and a more realistic solution.

When comparing this new CNES-CLS22 MDT with independent height observations from T/S profiles, we observe a global improvement in the CNES-CLS22 MDT compared with CNES-CLS18, but this is not true in all regions. South of the Atlantic and Indian Oceans (up to Australia), the variance of differences is reduced for CNES-CLS22 by more than 10% in many cells. Areas where the variance of differences with observations is increased are concentrated in the north-western Atlantic Ocean, particularly south of Greenland, in the far north of the Pacific Ocean (in the Gulf of Alaska, and close to Russia) and in the southern Pacific, where an increase in the variance of differences of 10% is observed.

Globally, the two solutions, CNES-CLS22 and CNES-CLS18, were close considering independent drifter comparisons. The comparison of the two solutions showed that CNES-CLS22 was the best, but by less than 1%.

Regional Black Sea MDT CMEMS_2020_BLK

The MDT CMEMS_2020_BLK is fairly smooth and does not represent mesoscale structures. Compared to independent heights observations estimated from T/S profiles, the MDT CMEMS_2020_BLK was better than the two others in the centre of the basin. In the West, it can be noted that the CMEMS MFC MDT (model MDT) performed better in areas close to the coast. For both velocity components, the MDT_CMEMS_2020_BLK showed better agreement with the independent drifters than the other MDTs compared.

Regional Mediterranean Sea MDT MDT_CMEMS_2020_MED

The MDT MDT_CMEMS_2020_MED represents mesoscle structures. This MDT was compared with independent in-situ measurements from drifters and from profiles T/S; these comparisons were performed by zone. In all zones, Pearson correlation coefficients are higher for the new MDT than for the previous one and the RMSD (Root Mean Squared Difference) is lower both for the height and for the geostrohic current.

Regional European MDT MDT_CMEMS_2024_EUR

The spectral analyse of the CMEMS_2024_EUR MDT showed that this MDT reached an energy plateau around 20 km. Analysis of filtered MDT at 20 km showed that scales below 20 km are noisy. In terms of current amplitude, MDT CMEMS_2024_EUR is better than CNES-CLS22 in the northern part of the zone; in the western part of the zone there is no clear improvement. Most of the cells show that CMEMS_2024_EUR is better than CNES-CLS22. In terms of current direction, CMEMS_2024_EUR showed improvements in the north, while CNES-CLS22 performed better in the west area of the Iberian Peninsula.

I.3 Estimated Accuracy Numbers

Mean Geostrophic current (associated with MDT):

EAN on mean geostrophic current are deduced from comparison between altimeter geostrophic velocities obtained using MDT solution (ADT associated geostrophic current) and independent drifter measurements corrected from ageostrophic signals. A synthesis is presented in Table 5.

Table 5: RMS of the differences between the zonal component U and the meridional component V of the current with the independant drifting buoy velocities (independent from MDT solution evaluated) and the altimeter geostrophic velocities obtained using MDT solution.

Product	Area	Zonal U [cm/s]	Meridional V [cm/s]
CNES-CLS22 MDT	Global (up to 70°N)	10.57	10.62
MDT_CMEMS_2020_BLK	Black Sea	11.50	11.20
MDT_CMEMS_2020_MED	Mediterranean Sea	10.38	11.47
CNES-CLS22 MDT	European coasts	8.91	8.89
CMEMS_Europe_2024 MDT	European coasts	8.89	8.89

II

II PRODUCTION SYSTEM DESCRIPTION

II.1 Production center name

SL-CLS-TOULOUSE-FR

II.2 Operational system name

DUACS

II.3 Production centre description

II.3.1 Key definitions

Sea Surface Height (SSH): is the height of the sea surface above the reference ellipsoid and it is measured as follows:

SSH = Satellite Altitude - Altimeter Range - Corrections

Where the 'Altimeter Range', measured by the altimeter, is the distance between the center of mass of the satellite to the surface of the Earth (see Figure 1); the 'Satellite Altitude' refers to the distance of the center of mass of the satellite above a reference point. The reference point will usually be either on the reference ellipsoid or the center of the Earth. 'Corrections' due to environmental conditions need to be applied in order to retrieve the correct 'Sea Surface Height'. They are listed in the SLA QUID[1] ().

Geoid: is a gravity equipotential surface that would correspond with the ocean surface if ocean was at rest (i.e. with no currents under only the gravity field). Then, when the ocean is also influenced by wind, differential heating and precipitation and other sources of energy, the ocean surface moves from the geoid.

Mean Sea Surface (MSS_N): is the temporal mean of the SSH over a period N. It is a mean surface above the ellipsoid of reference and it includes the geoid.

$$MSS_N = \langle SSH \rangle_N$$

Although the reference period N considered can be changed as described in Pujol et al. (2018), it is currently 1993-2012.

Sea Level Anomaly (SLA_N): is the anomaly of the signal around the time-mean component. It is deduced from the SSH and MSS_N :

$$SLA_N = SSH - MSS_N$$

[1] https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SL-QUID-008-032-051.pdf

Mean Dynamic Topography (MDT_N): is the temporal mean of the SSH above the geoid over a period N.

$$MDT_N = MSS_N - geoid$$

The departure from the geoid (i.e., MDT) provides information on the mean effect on SSH from ocean dynamics.

Absolute Dynamic Topography (ADT): is the instantaneous height above the geoid and it is measured as the sum of the SLA_N and MDT_N :

$$ADT = SLA_N + MDT_N = SSH - MSS_N + MDT_N$$

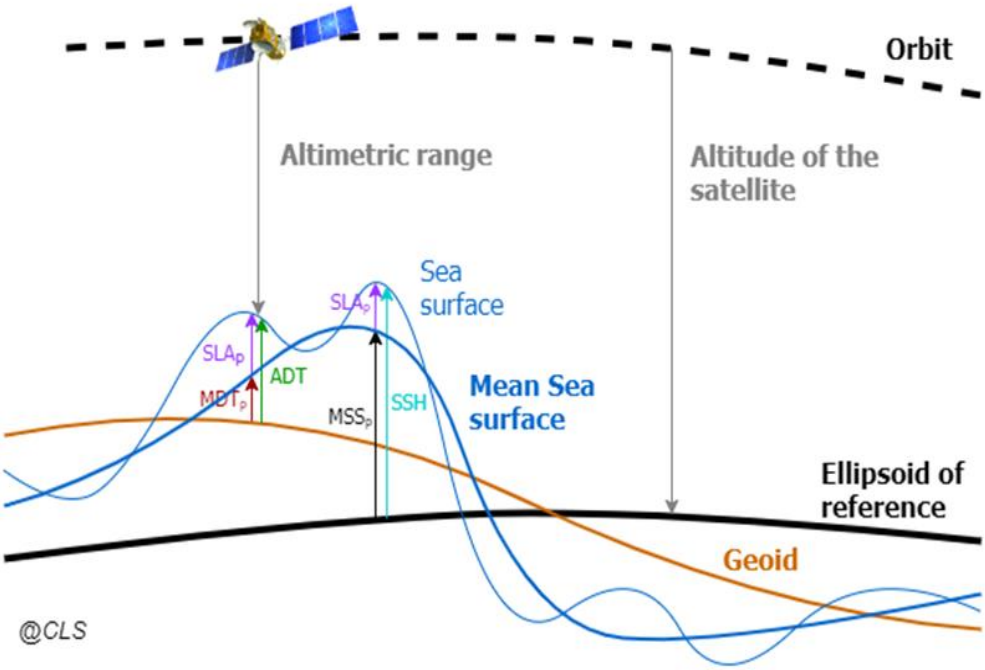

Figure 1: Diagram of the various quantities indicating the relative reference to which they are measured. SSH: Sea Surface Height; SLA: Sea Level Anomaly; MDT: Mean Dynamic Topography; ADT: Absolute Dynamic Topography; MSS: Mean Sea Surface; $_p$: period over which the mean values are calculated (Credits: CLS).

II.3.2 Mean Dynamic Topography computation methodology

The method used to calculate the Mean Dynamic Topography (MDT) is similar to the one used in previous CNES-CLS MDT versions. A detailed description can be found in Rio and Hernandez (2004); Rio et al. (2011; 2014a). It is a three step approach (see sections below for details).

First step: computation of the first guess

In a first step, a first guess MDT solution is calculated from the optimally filtered differences between an altimeter Mean Sea Surface (MSS) and a geoid model. The effective resolution of the obtained field depends on the level of noise of the raw difference between MSS and geoid height. It thus depends on the areas.

For the MDT CNES-CLS22, the large scale first-guess MDT is computed by filtering at 125 km (Bruinsma et al. 2014) the difference between the CNES_CLS22 MSS (Schaeffer et al. 2023) and the GOCO06S (Kvas et al. 2021) geoid height. For the MDT CNES-CLS22 computation, this step has been improved with the application of additional Lagrangian filter along the coast to avoid streamline going into land.

The attenuation of the normal speed at the coast is shown in Figure 2, for the Mid Atlantic Bight, coastal region running from Massachusetts to North Carolina (left), and the global area (right). When the attenuation is between 0 and 0.5, the currents perpendicular to the coast are well attenuated. When the attenuation is close to 1, we have not necessarily attenuated the currents, even we have amplified them a little. Lagrangian filtering tends to spread out this strong current, as can be seen in Figure 2 (left) with the Gulf stream around 36°N. Globally, these normal currents at the coast are attenuated (Figure 2, right).

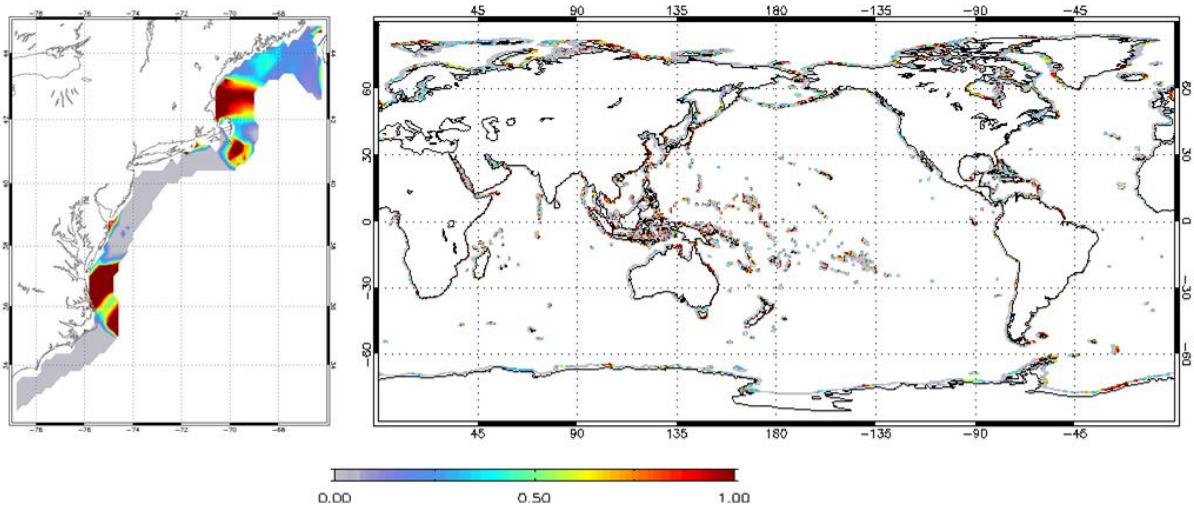

Figure 2: Lagrangian attenuation of normal velocity near coast. Left: Mid Atlantic Bight (coastal region running from Massachusetts to North Carolina); Right: Global area.

For regional MDTs over the Black Sea and Mediterranean Sea, large scale first-guess MDT is computed by filtering the difference between the CNES-CLS15 MSS (Pujol et al. 2018) and the geoid height from Eigen6c4 (Shako et al. 2014) at 100 km resolution with a 2d Gaussian filter. For the Mediterranean Sea MDT, the main difference with SMDT 2014 is that the first guess is made from altimetry data and a geoid model (MSS- geoid) instead of a first guess made from a numerical model.

Second step : estimation of the synthetic fields

In the second step of the method, synthetic estimates of the MDT and mean geostrophic velocities are calculated using in-situ measurements of the ocean dynamic heights and surface velocities. First the in-situ measurements are processed so as to extract the geostrophic component only from the drifting buoy total velocities, and to complete the dynamic heights with the missing barotropic and deep baroclinic components. The temporal variability of the measured heights and velocities is further removed by subtracting the altimeter sea level and geostrophic velocity anomalies respectively. The processed in-situ measurements are further averaged into cells to obtain respectively the synthetic mean heights and velocities.

With the CNES-CLS2018 MDT, this step was improved, drifter processing in particular (Ekman model and wind slippage, Mulet et al. 2021). For this new CNES-CLS2022 version, the Ekman model has been further improved by the use of the ERA5 wind model (instead of ERAinterim). Further improvements are the refinement of the Ekman model in latitude resolution by Mixed Layer Depth (method detailed in CMEMS-MOB-QUID-015-004[2]). Wind slippage, while retaining the same calculation method as in older versions, has also evolved (specifically the impact of the change in the Ekman model and the new processing af altimetry data).

For the estimation of mean synthetic heights, the methodology has been maintained, but the implementation has been improved, which has enabled us to gain points close to the coast.

Specific processing in the Arctic

In the Arctic, SLA maps are not defined everywhere (globally, we have no data further north than 80°N). In-situ data processing is therefore degraded, as it is impossible to reduce temporal variability. For the CNES-CLS2022 MDT, these degraded data were kept for the MDT calculation, but the error associated with them was increased.

Specific processing in the Black Sea

Note that the observed dynamic heigth was not used for the regional Black Sea MDT computation (see Table 8).

Third step : multivariate objective analysis

These synthetic fields are finally used in the third step to improve the accuracy of the filtered MDT obtained at step 1 and bring information at shorter scales. This is done through a multivariate objective analysis whose required inputs are: (i) the synthetic mean heights and velocities and their errors, (ii) the first guess MDT, (iii) the a-priori knowledge of the MDT variance and (iv) the zonal and meridional correlation scales.

For the regional Black Sea MDT, this processing was improved applying a double-step objective analysis. In the first objective analysis, a large correlation radius (300 km) is used to improve large scales. In the second analysis, this radius is reduced to 150 km to improve smaller scales. Aditionally, regional MDTs are computed with a finer resolution up to 1/24°, compared to the Global solution.

II.3.3 Standards and improvements of differents products

The Table 6 to Table 9 summarize the standards used respectively for the Global MDT CNES_CLS22, regional MDTs for the Mediterranean (MDT_CMEMS_2020_MED) and the Black Sea (MDT_CMEMS_2020_BLK), and for European facade (MDT_CMEMS_2024_EUR). When existing, they also show the differences with previous MDTs versions.

[2] https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-MOB-QUID-015-004.pdf

Table 6: Standards and improvements done for the computation of the most recent MDT CNES-CLS22 version compared with the previous MDT CNES-CLS18 version. Improvements over the previous version are marked in bold.

	MDT CNES-CLS18	**MDT CNES-CLS22**
MSS	CNES-CLS15 (Pujol et al. 2018)	**CNES-CLS22** (Schaeffer et al. 2023)
Geoid	GOCO05S (Mayer-Gürr 2015) Complete GOCE mission (Nov 2009-October 2013) + 10.5 years of GRACE data	**GOCO06S** (Kvas et al. 2021) Complete GOCE mission fully reprocessed (Nov 2009-October 2013) + 14 years of GRACE data
First Guess	Observed first guess (MSS – geoid)	Observed first guess (MSS – geoid)
First Guess filtering	Optimal filter (Rio et al. 2011) with updated parameters	Optimal filter (Rio et al. 2011) **+ lagrangian filter along the coast to avoid streamline going into land**
Hydrological data	CTD and ARGO Pref variable 200/400/900/1200/1900 from CORA4.2 (1993-2013), CORA5.0 (2014-2015) and CORA5.1 (2016) Period 1993-2016	CTD and ARGO Pref variable 200/400/900/1200/1900 from CMEMS CORA Release 202212 Period **1993-2020**
Drifter Data	SD-DAC drifter, both drogued and undrogued: 1993-2016 Argo floats surface velocities: 1997-2016	SD-DAC drifter, both drogued and undrogued: **1993-2021** Argo floats surface velocities: **1997-2021**
High Frequency Radar Data	-	**2006-2016 Mean HF radar currents processed by Rutgers University** (Roarty et al. 2020) in the Mid Atlantic Bight
Ekman model	Preliminary version of the Copernicus-Globcurrent Ekman model (MULTIOBS_GLO_PHY_NRT_015_003 and MULTIOBS_GLO_PHY_REP_015_004) Parameters fitted over the period 1993-2016 by latitude and Mixed Layer Depth (from CMEMS ARMOR3D: MULTIOBS_GLO_PHY_REP_015_002) Two levels: 0 m and 15 m	Copernicus-Globcurrent Ekman model (MULTIOBS_GLO_PHY_REP_015_004), used **ERA5 wind fields**. Parameters fitted over the period **1993-2017 by latitude and Mixed Layer Depth at finer resolution** (from CMEMS ARMOR3D: MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012) Two levels: 0m and 15m
Wind Slippage correction	Update of (Rio, Guinehut, and Larnicol 2011) in order not to discard the trajectories beginning/end	Same version of the method but with an improved Ekman model, and new processing of altimetry data.
Drifter filtering	Max (24 hours, Inertial Period)	Max (24 hours, Inertial Period)
Altimeter data	Delayed-Time CMEMS-DUACS 2018 (Taburet et al. 2019)	Delayed-Time **CMEMS-DUACS 2021** (Taburet et al. 2022)

Table 7: Standards and improvement done for the regional MDT CMEMS_2020_MED computation compared with the previous SMDT_2014 (Rio et al. 2014b). Improvements over the previous version are marked in bold.

	SMDT 2014 (Rio et al. 2014b)	**MDT_CMEMS_2020_MED**
MSS	-	**CNES-CLS15** (Pujol et al. 2018)
Geoid	-	**Eigen6c4** (Shako et al. 2014)
First Guess filtering	-	2d Gaussian filter R=100 km
First Guess	Modeled MDT computed averaging over the 1993–1999 period outputs from the 1/16° MFS model (Mediterranean Forecasting System, Adani et al. (2011)	**Observed first guess (MSS – geoid)**
Hydrological data	CTD and Hydrological profiles collected by IMEDEA (Mediterranean Institute for Advanced Studies), by SOCIB (Ruiz et al. 2009; 2012; Bouffard et al. 2010; Pascual et al. 2013; Heslop et al. 2012) and by IEO (Spanish Institute of Oceanography; IBAMar database, López-Jurado et al. 2005; Alemany et al. 2010). Argo floats and CTD measurements from the EN3 database. Period 1993-2012	CTD and ARGO Pref variable 350 from CORA4.2 (1993-2013), CORA5.0 (2014-2015) and CORA5.1 (2016) Period **1993-2018**
Drifter Data	Surface drifting buoy (drifter), both drogued and undrogued (OGS dataset): 1993-2011 Drifters over June-December 2011 kept for validation	Surface drifting buoy (drifter), both drogued and undrogued: **1993-2016** (Menna et al. 2017) DOI:10.6092/7a8499bc-c5ee-472c-b8b5-03523d1e73e9 **15% of drifters kept for validation**
Ekman model + wind Slippage > wind driven currents	Method and parameters: Poulain et al. (2012) Wind : Cross-calibrated, multi-platform (CCMP) ocean surface wind velocities (Atlas et al. 2011)	Method and parameters: Poulain et al. (2012) Wind : Cross-calibrated, multi-platform (CCMP) ocean surface wind velocities (Atlas et al. 2011)
Drifter filtering	36 h	36h
Synthetic mean heights and velocities	1/8° synthetic mean velocities 1/4° synthetic mean heights	1/8° synthetic mean velocities 1/4° synthetic mean heights
Altimeter data	Delayed-Time DUACS-2010 (Dibarboure et al. 2011)	Delayed-Time **CMEMS-DUACS 2018** (Taburet et al. 2019)
Resolution	Global 1/8°	Global **1/24°**

Table 8: Standards used for the regional MDT CMEMS_2020_BLK computation.

	MDT_CMEMS_2020_BLK
MSS	CNES-CLS15 (Pujol et al. 2018)
Geoid	Eigen6c4 (Shako et al. 2014)
First Guess	Observed first guess (MSS – geoid)
First Guess filtering	2d Gaussian filter R=100km
Hydrological data	-
Drifter Data	Surface drifting buoy (drifter), both drogued and undrogued: 1999-2009 (Menna et al. 2017) DOI:10.6092/B40CD642-9555-44FA-8B91-3CD88B6C225B 20% of drifters kept for validation
Ekman model + wind Slippage > wind driven currents	Method : linear regression model (Poulain et al. 2012) Parameters : (Menna and Poulain 2014) Wind : Cross-calibrated, multi-platform (CCMP) ocean surface wind velocities (Atlas et al. 2011)
Drifter filtering	24 hours (Inertial Period about 17h)
Synthetic mean heights and velocities	1/8° synthetic mean velocities no synthetic mean heights
Altimeter data	Delayed-Time CMEMS-DUACS 2018 (Taburet et al. 2019)
Resolution	1/16°

Table 9: Standards and improvement done for the MDT_CMEMS_2024_EUR computation compared with the previous MDT CNES-CLS22. Improvements over the previous version are marked in bold.

	MDT CNES-CLS22	**MDT_CMEMS_2024_EUR**
MSS	CNES-CLS22 (Schaeffer et al. 2023)	CNES-CLS22 (Schaeffer et al. 2023)
Geoid	GOCO06S (Kvas et al. 2021) Complete GOCE mission fully reprocessed (Nov 2009-October 2013) + 14 years of GRACE data	GOCO06S (Kvas et al. 2021) Complete GOCE mission fully reprocessed (Nov 2009-October 2013) + 14 years of GRACE data
First Guess	Observed first guess (MSS – geoid)	Observed first guess (MSS – geoid)
First Guess filtering	Optimal filter (Rio et al. 2011) + lagrangian filter along the coast to avoid streamline going into land	Optimal filter (Rio et al. 2011) + lagrangian filter along the coast to avoid streamline going into land
Hydrological data	CTD and ARGO Pref variable 200/400/900/1200/1900 from CMEMS CORA Release 202212 Period 1993-2020	CTD and ARGO Pref variable 200/400/900/1200/1900 from CMEMS CORA Release 202212 Period 1993-2020
Drifter Data	SD-DAC drifter, both drogued and undrogued: 1993-2021 Argo floats surface velocities: 1997-2021	SD-DAC drifter, both drogued and undrogued: **1993-2023** Argo floats surface velocities: **1997-2021**
High Frequency Radar Data	2006-2016 Mean HF radar currents processed by Rutgers University (Roarty et al. 2020) in Mid Atlantic Bight	**2009-2017 Mean HF radar currents** processed by Caballero et al. (2020) in the Bay of Biscay
Ekman model	Copernicus-Globcurrent Ekman model (MULTIOBS_GLO_PHY_REP_015_004), used ERA5 wind fields. Parameters fitted over the period 1993-2017 by latitude and Mixed Layer Depth at finer resolution (from CMEMS ARMOR3D: MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012) Two levels: 0 m and 15 m	Copernicus-Globcurrent Ekman model (MULTIOBS_GLO_PHY_REP_015_004), used ERA5 wind fields. Parameters fitted over the period **1993-2020** by latitude and Mixed Layer Depth at finer resolution (from CMEMS ARMOR3D: MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012) Two levels: 0 m and 15 m
Wind Slippage correction	Same version of the method but with an improved Ekman model, and new processing of altimetry data.	Same version of the method but with an improved Ekman model, and new processing of altimetry data (from CMEMS INSITU_GLO_PHY_UV_DISCRETE_MY_013_044)
Drifter filtering	Max (24 hours, Inertial Period)	Max (24 hours, Inertial Period)
Altimeter data	Delayed-Time CMEMS-DUACS 2021 (Taburet et al. 2022)	Delayed-Time CMEMS-DUACS 2021 (Taburet et al. 2022)
Resolution	1/8°	**1/16°**

III VALIDATION FRAMEWORK

The MDT_HYBRID_CNES_CLS22_CMEMS2020 is the merge of the MDT CNES-CLS22 (Jousset et al. 2022; 2023b) with the regional MDT CMEMS_2020_MED and MDT CMEMS_2020_BLK (Jousset et al. 2023a) interpolated on a global MDT grid. Validation results for the MDT_CNES_CLS22 are described in this section (IV.1). Firstly, the MDT was validated in height with a spectral analysis in comparison with other MDT solutions, and compared with independent observations (mean bias, reduction of variance, RMS, Pearson correlation coefficient). Secondly, the associated geostrophic currents of the MDT were validated with a comparison to independent drifter observations (mean bias, RMS, Pearson correlation coefficient).

Validation results for the Black Sea (CMEMS_2020_BLK MDT, Jousset et al. 2023a) are described in section IV.2, for the heigth and for the geostrophic currents associated. The validation consisted in comparisons with independent in-situ measurements from drifters (mean bias, correlation coefficient, RMS of differences) and the MDT was also compared to others existing MDTs on the area.

Validation results for the Mediterranean Sea (CMEMS_2020_MED MDT, Jousset et al. 2023a) are described in section IV.3. The validation consisted in the comparison with independent in-situ measurements from drifters and from profiles T/S. The MDT_ CMEMS_2020_MED was also compared to others existing MDTs.

Validation results for the European coasts MDT are described in section IV.4. The validation consisted in a comparison in heigth with others existing MDTs and with independent drifters for geostrophic currents associated.

IV VALIDATION RESULTS

IV.1 Global MDT_HYBRID_CNES_CLS22_CMEMS2020

IV.1.1 Mean Dynamic Topography height

Validation when comparing through a spectral analysis

Different MDT solutions were compared through a spectral analysis:

- first guess (called FirstGuess22)
- previous CNES-CLS13 MDT
- previous CNES-CLS18 MDT
- new CNES-CLS22 MDT
- Glorys12 numerical model MDT (1/12° numerical model from Mercator-Ocean and distributed within CMEMS: product GLOBAL_REANALYSIS_PHY_001_030[3]

An example is given in Figure 3 for a 10° x 10° cell in the Antarctic Circumpolar Current area. According to this Figure, as for the CNES-CLS18 MDT (in blue), the CNES-CLS22 MDT (in red) increases energy at short scales compared to either the first guess (based on GOCE data, in purple) and the previous CNES-CLS13 MDT (in green). The energy level is in good agreement with the GLORYS12 energy level (in black) until 50 km wavelength (25 km of resolution). For the CNES-CLS18 MDT, at shorter scales a flat response is obtained; this means that the shortest scales (between 12.5 km and 25 km of resolution) in the CNES-CLS18 MDT might be contaminated by noise. For CNES-CLS22 MDT, the curve is more in line with Glorys than CNES-CLS18 MDT. At small scales the curve continues decreasing, probably because the new version is less affected by noise.

[3] Global Ocean Physics Reanalysis | Copernicus Marine Service

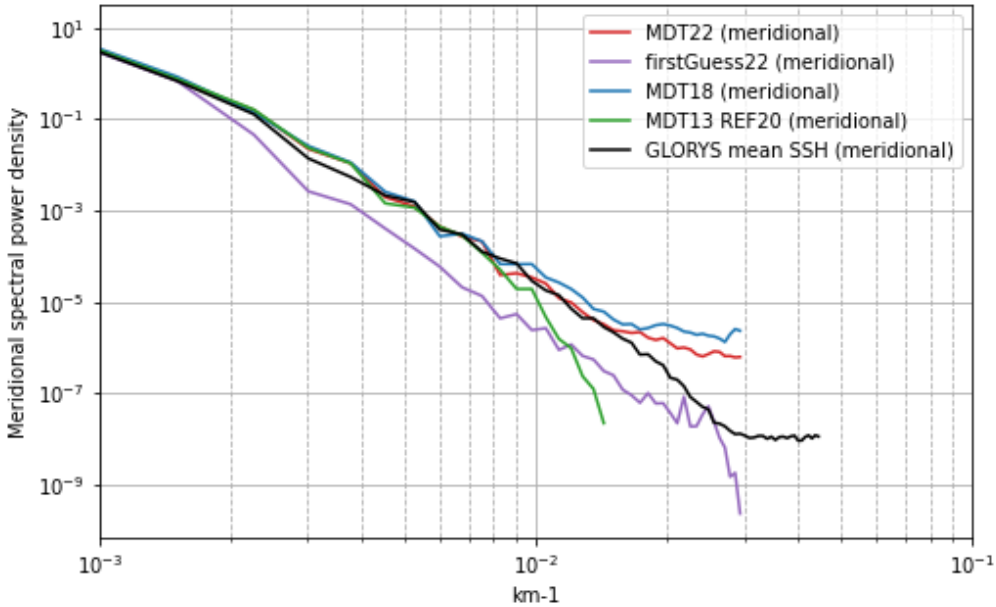

Figure 3: Spectral Power Density obtained for different MDT solutions in a 10° x 10° cell in the Antarctic Circumpolar Current area for the meridional current. MDT CNES-CLS22 First Guess (purple); CNES-CLS18 MDT (blue); CNES-CLS22 MDT (red); CNES-CLS13 MDT (green); GLORYS12 (black).

Validation when comparing with independent data

The CNES-CLS18 and CNES-CLS22 solutions were also compared with independent data. Independent data for CNES-CLS22 and CNES-CLS18 MDTs were randomly selected from 2017 (around 5% of the T/S profile datasets) and kept for validation. The dynamic height estimated from T/S profiles (from a reference depth of 200 m, 400 m, 900 m, 1200 m or 1900 m, depending on the depth of the profile in question) characterizes a baroclinic component of the dynamic circulation. Whereas altimetry measures a height that is also influenced by baroclinic processes occurring from the reference depth to the bottom of the water column, as well as by barotropic processes. For validation purposes, we choose to keep only the deepest profiles (with a reference depth of 1900 m), thus reducing the number of deep baroclinic processes not taken into account. Under these conditions, we got a validation set comprising 2% of the database.

As a first step, we compared the CNES-CLS18 and CNES-CLS22 MDTs against these independent dynamic heights, by looking at the correlation of the ADT (SLA+MDT considered) and the independent dynamic heights (Figure 4, (a) CNES-CLS18 and (b) CNES-CLS22). Correlations (Pearson correlation) are calculated in cells of 5° x 5° (with at least 20 data). Correlation coefficients are high, mostly between 0.8 and 1 for both MDTs, and the different MDTs have equal correlations.

Secondly, Figure 4 (c) shows the mean bias per 5° x 5° cell between the ADT estimated from the CNES-CLS22 MDT and the independent dynamic heights. This bias is significant (globally mean bias of 1.30 m), with spatial variations in the Norwegian Sea and in the south near Antarctica, and it mainly represents the barotropic component not observed by the dynamic heights. This bias is similar for CNES-CLS18, as the differences between the two MDTs is of the order of 10 cm at most, compared with the larger globally mean bias (1.30 m). This is why Figure 4 (d) shows a comparison between the variances of the differences (not considering the bias) between the different ADTs and the dynamic

heights, in percent. In blue the variance of differences is reduced (whereas in red, the variance of differences increases) using CNES-CLS22 compared with CNES-CLS18.

Globally, we observe an improvement in CNES-CLS22 MDT compared with CNES-CLS18, but this is not true in all regions. South of the Atlantic Ocean and the Indian Ocean (as far south as Australia), the variance of the differences is reduced for CNES-CLS22 by more than 10% in many cells (even if cells of strong reduction are juxtaposed with cells of increased variance of the differences). Areas of degradation (i.e., cells where the variance of the differences with observations is increased) are concentrated in the north-western Atlantic Ocean, particularly south of Greenland, in the very north of the Pacific Ocean (along the Gulf of Alaska to the Fox Islands, and close to Russia) and in the south of the Pacific, where there are degradations of over 10% (also juxtaposed with cells where the variance is reduced).

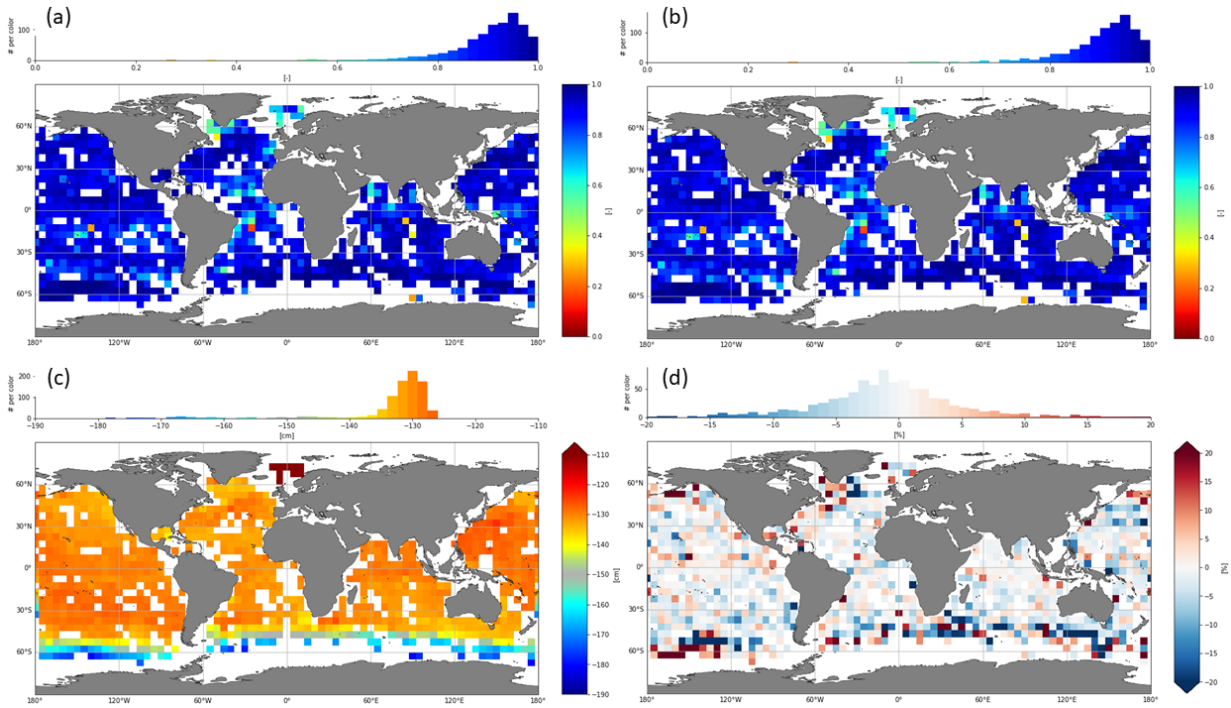

Figure 4: Correlation (Pearson correlation coefficient) between (a) dynamic heights estimated from independent T/S profiles and ADT calculated from CNES-CLS18 MDT and (b) from CNES-CLS22 MDT. (c) shows the bias between the dynamic heights estimated from the independent T/S profiles and the ADT calculated from the CNES-CLS2 MDT. And (d) shows the reduction/increase in variance of the differences between independent dynamic heights and ADT (from CNES-CLS18 and CNES-CLS22) as a percentage. In blue , the variance of the differences is reduced (whereas in red, the variance of differences increases) when using CNES-CLS22 compared with CNES-CLS18. All these statistics were calculated in 5° x 5° cells. Only cells with at least 20 data are kept.

Quantitative regional validation in the Arctic

Figure 5 shows the (a) CNES-CLS18 MDT; (b) CNES-CLS22 MDT in the Arctic, and (c) shows the differences between them (measured as CNES-CLS22 MDT - CNES-CLS18 MDT). In contrast to the CNES-CLS18 MDT, the CNES-CLS22 MDT covers the Arctic zone thanks to the improved coverage of the CNES-CLS22 MSS used to estimate the first guess of this new MDT. Figure 5 (d) and (e) show the mean

synthetic heights used as input data for CNES-CLS18 MDT and CNES-CLS22 MDT respectively. With this new version, we took into account data for the entire area covered by the MDT.

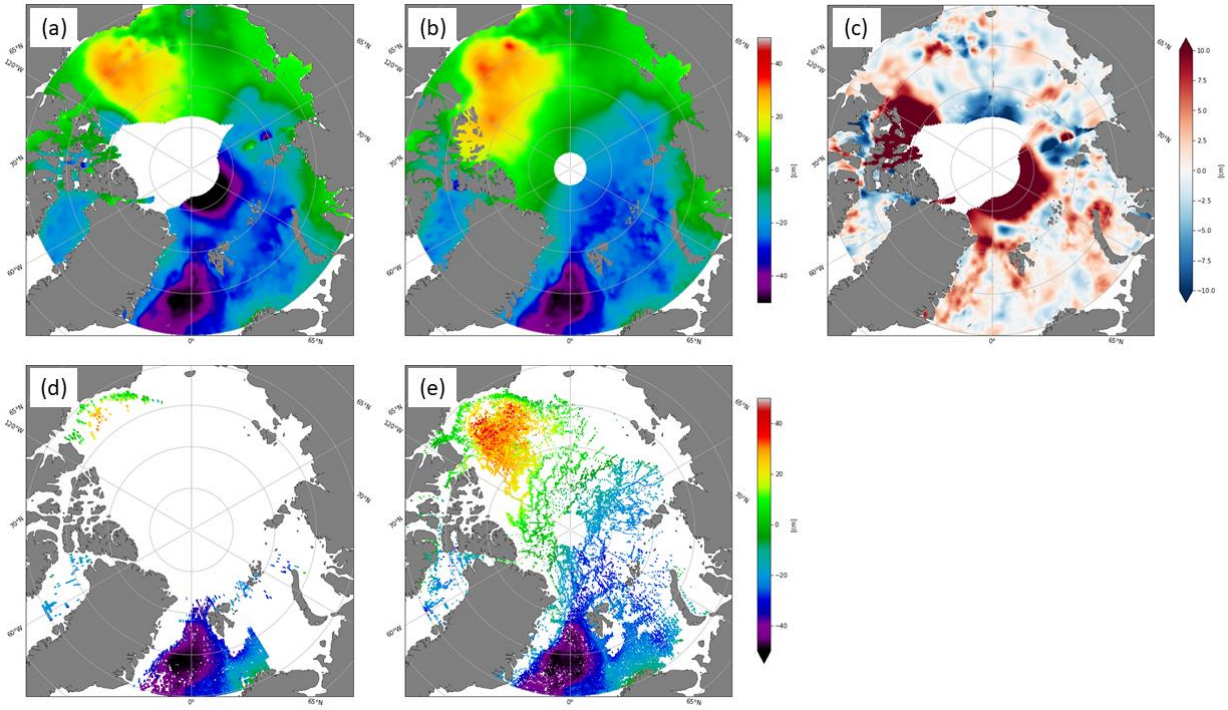

Figure 5: Quantitative regional validation in the Arctic. (a) CNES-CLS18 MDT and (b) CNES-CLS22 MDT in Arctic. (c) Differences between them (measured as CNES-CLS22 MDT - CNES-CLS18 MDT). Mean synthetic heights for input data for (d) CNES-CLS18 MDT and (e) CNES-CLS22 MDT.

Artifacts present on the CNES-CLS18 MDT have disappeared from the new version, for example, around 110°E. Figure 6 shows a zoom around the island of Svalbard. The white line representing the 200 m isobath makes it possible to locate the Yermak plateau. The new MDT has contours that follow this bathymetry much better than the CNES-CLS18 MDT, this improves the circulation shown in Figure 6 (a) (Meyer et al. 2017).

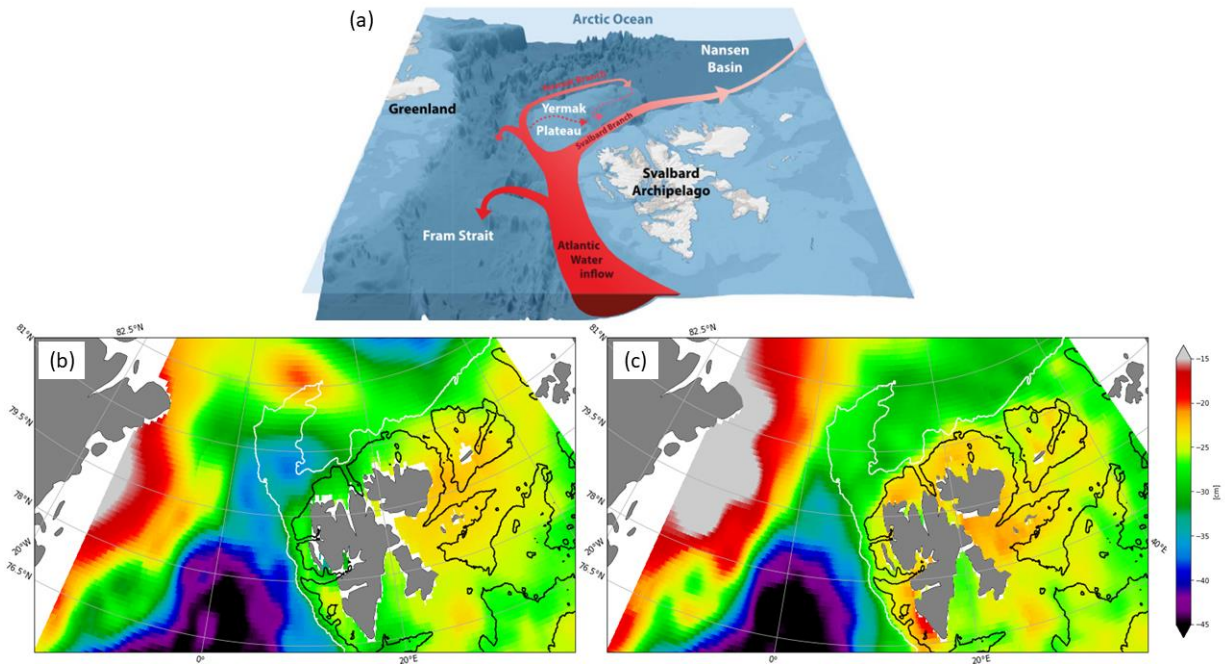

Figure 6: Quantitative regional validation in the Arctic. (a) Regional circulation schematic of Atlantic Water inflow in Fram Strait and Nansen Basin, west and north of the Svalbard Archipelago (Meyer et al. 2017); and (b) CNES-CLS18 MDT and (c) CNES-CLS22 MDT in the area. The black line represents the 1000 m isobath and the white line the 200 m isobath.

IV.1.2 Mean Geostrophic Currents

Validation when comparing with independent velocity data

The CNES-CLS18 and CNES-CLS22 solutions were compared with independent velocity data. From the AOML drifters dataset, 10% subset of the sample was randomly selected and kept for use in the validation as additional independent data. Figure 7 shows the drifters kept for the validation of the MDT. Figure 7 (a) shows the number of measurement points of independent drifters kept for validation per ¼° x ¼° cell and (b) shows the modulus of mean current per ¼° x ¼° cells. There are few drifters close to Antarctica (south of about 50°S), at the equator and in the Arctic. In addition, the Atlantic is slightly better sampled than the other basins, and in particular the North Indian.

To compare with the MDTs, we removed the wind-driven current (Ekman current and wind slippage for undrogued drifters) from the total drifters current. Then, data were filtered at inertia frequency (if inertia frequency was between 1 and 5 days, otherwise a minimum of 1 day and a maximum of 5 days was taken) to remove the tide and inertial waves. The objective is to keep only the geostrophic signal. As the geostrophy hypothesis is no longer valid at the equator, we exclude drifters between 5°S and 5°N at the equator.

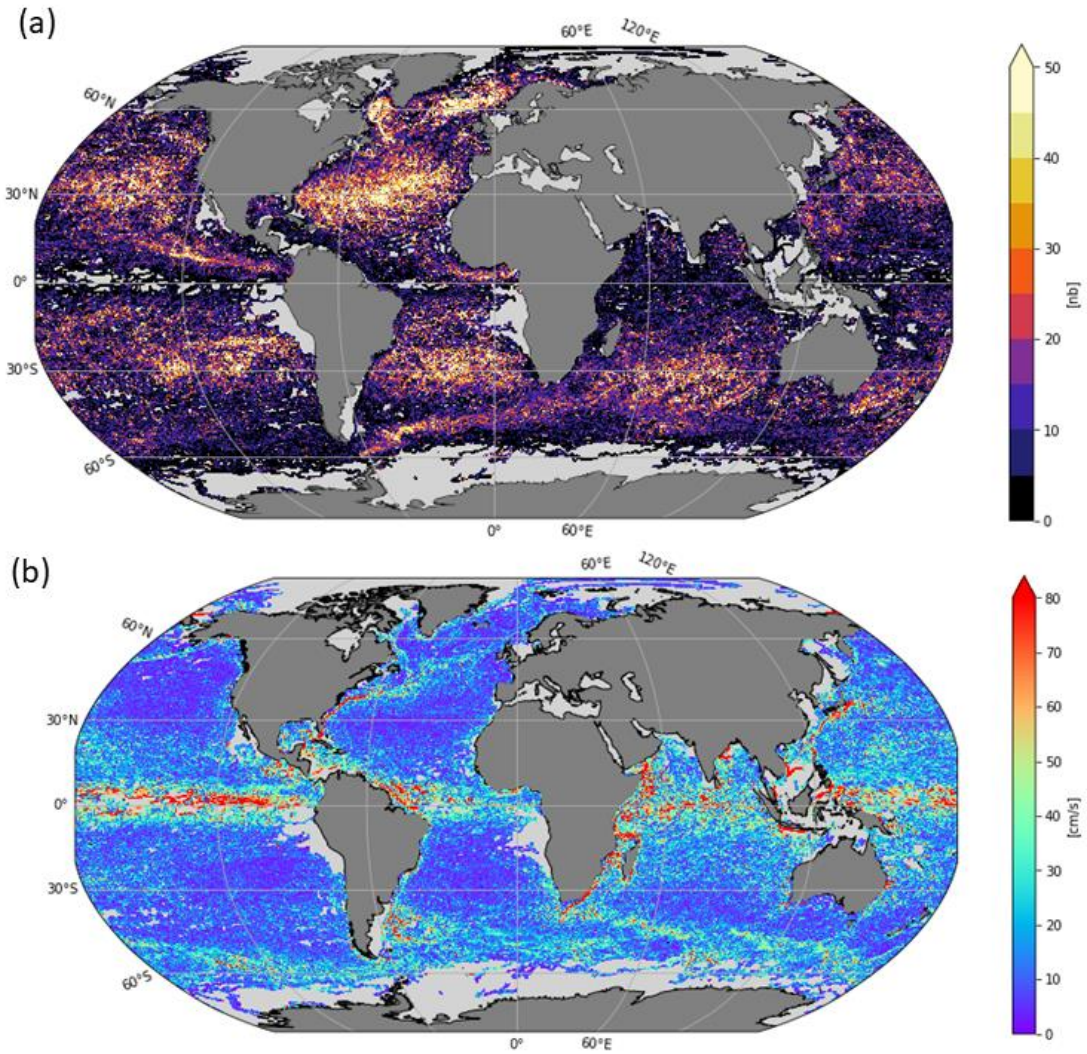

Figure 7: Independent drifters kept from the AOML database for validation: (a) Number of points of size 0.25° x 0.25° cells and (b) Modulus of the mean current in 0.25° x 0.25° cells.

Absolute dynamic topography values were calculated by adding the CMEMS gridded SLA to the new CNES-CLS22 MDT. Associated geostrophic current were then derived and interpolated along the drifter trajectories. Root Mean Square differences (RMSD) between the obtained geostrophic velocities and the drifter derived geostrophic velocities were then calculated. The same comparison was performed using the previous CNES-CLS18 MDT. The percent of RMSD to the independent velocity dataset (divided by the Root Mean Square of the drifters signal) were then calculated as a function of distance to the coast (Figure 8). In this Figure, results for the CNES-CLS22 and for the CNES-CLS18 are shown in blue and in green respectively, for the zonal (Figure 8a) and meridional (Figure 8b) components, and also for modulus (Figure 8c) and direction (Figure 8d) of the current.

For the zonal and meridional components of the current, the RMSD in the MDT CNES-CLS22 is sligthly reduced everywhere compared with the CNES-CLS18 MDT (improvement about 1%, at distance less than 50 km). This diagnostic is also calculated in terms of current modulus and direction. Current

amplitude is slightly improved, but this is not necessarily the case for direction and for cells between 40 and 50 km, nor for those covering 70-80 km and 80-90 km. The CNES-CLS18 MDT's improvement over the old version took place close to the coast, while the new CNES-CLS22 MDT improved equally as the CNES-CLS18, or even slightly better, in the 0-50 km band from the coast, in terms of U, V and current amplitude.

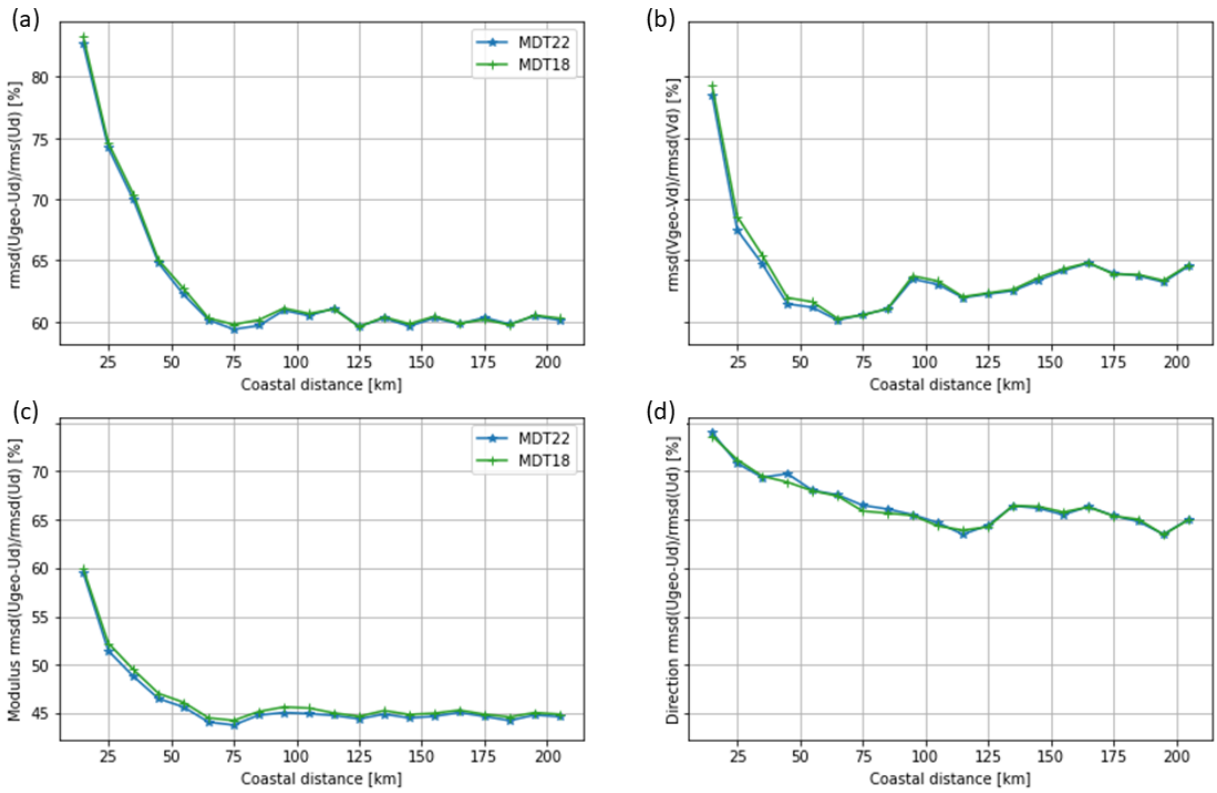

Figure 8: RMS differences of (a) zonal, (b) meridional, (c) modulus and (d) direction as a function of coastal distance between the independent drifting buoy velocities and the altimeter geostrophic velocities obtained using different MDT solutions: Blue the new CNES-CLS22 MDT and Green the CNES-CLS18 MDT. Statistics are calculated for the entire globe up to 70°N, excluding the equatorial band (5°S - 5°N).

Bias and Root Mean Square differences (RMSD) between the drifter derived geostrophic velocities and the geostrophic velocities obtained with CNES-CLS22 and CNES-CLS18 were also calculated spatially in 5° x 5° cells as shown in Figure 9. All these statistics were calculated with at least two different drifters and with at least 100 measurement points.

Comparisons of bias (a) in current modulus (in m/s) and (b) in direction (in degrees) are shown in the first row of Figure 9. In blue, there is a decrease (whereas in red there is an increase) of the bias of the geostrophic currents estimated with the CNES-CLS22 MDT and derived from the ADT. Using the new solution, globally there is a decrease in the bias in the modulus of the current (Figure 9a), with a greater decrease in the bias in areas with strong currents, such as the Gulf Stream (along the east coast of the United States), Kuroshio Current (along the Japanese coast), Agulhas Current (current along the east coast of South Africa, breaking away from the coast and returning eastwards) and Antarctic Circumpolar Current. The areas with the greatest degradation are South Greenland and

South Kerguelen. Regarding the directional bias (Figure 9b), the contribution of the new solution is less clear, as the direction of the geostrophic currents derived from the ADT using the CNES-CLS18 is better in the South between the Kerguelens and southern Australia. These are areas with fewer validation drifters.

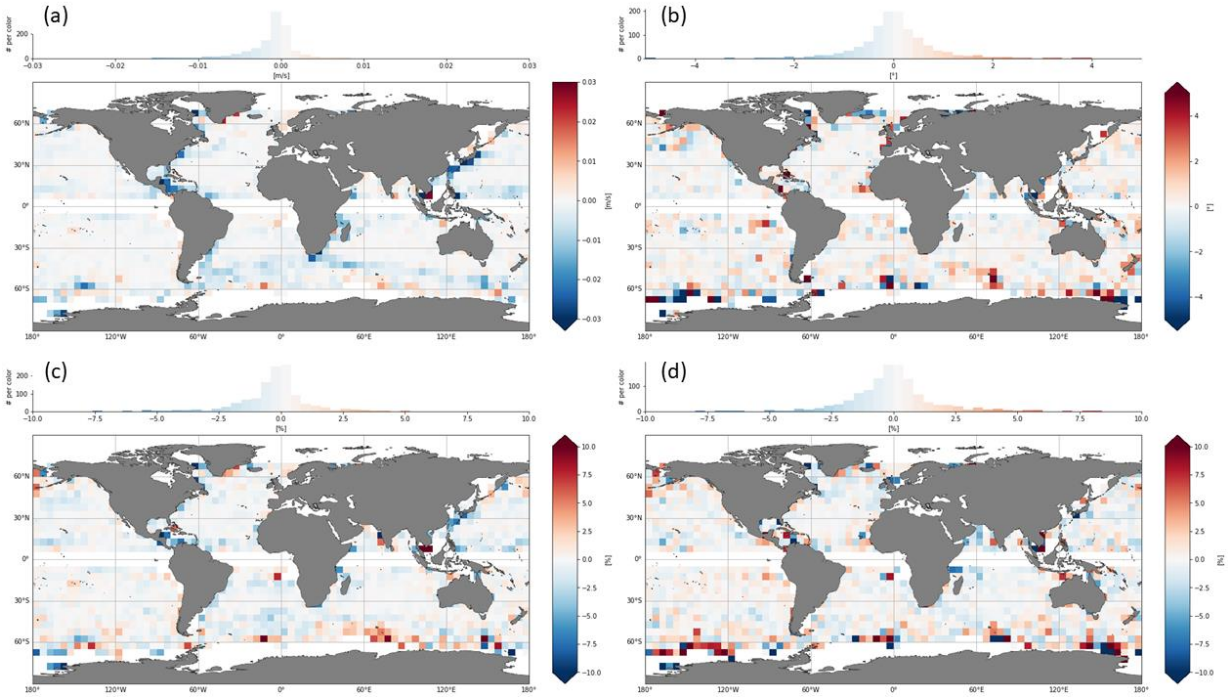

Figure 9: Comparison of bias in (a) current modulus and (b) current direction independent drifting buoy velocities with the altimeter geostrophic velocities obtained using different MDT solutions. Changes (decrease or increase) in the bias using CNES-CLS22 compared with CNES-CLS18 are shown. Comparison of the RMS of the differences (c) in current modulus and (d) in direction, of the independent drifting buoy velocities and the altimeter geostrophic velocities obtained using MDT solutions, CNES-CLS22 and CNES-CLS18, as a percentage, measured as: $\%RMSD = RMS(U_{ADT22}) - RMS(U_{ADT18} - U_d)/RMS(U_{ADT18} - U_d)$. All these statistics are calculated in 5° x 5° cells. Only cells with at least two different drifters and at least 100 measurement points were kept.

Comparisons of the RMS of the differences in (c) current modulus and (d) direction of the independent drifting buoy velocities and the altimeter geostrophic velocities obtained using the MDT solutions CNES-CLS22 and CNES-CLS18, as a percentage, are shown in the second row of Figure 9. In blue (respectively in red), the RMSD is reduced (respectively is increased) using CNES-CLS22 compared with CNES-CLS18. In terms of current amplitude, the majority of cells show an improvement of between 0 and 2.5% for the CNES-CLS22 solution, with in particulara greater improvement in the Kuroshio current (along the japanese coast) and in the Caribbean Sea. Areas of degradation in the current amplitude RMS are the area around Kerguélen (islands in the sub-Antarctic) in the Antarctic Circumpolar Current (ACC) and along the southeast coast of Greenland. Comparison of the RMS of the differences in the current direction showed a more mixed result, with degradation cells compared to the CNES-CLS18 solution, specifically in the ACC in the South Pacific. It must also be noted that some cells showed an improvement in the RMS of the differences in the CNES-CLS22 solution for the current amplitude but not for the direction, and vice versa. This is the case for some cells south of the Kerguélen Islands and at the extreme south of the Pacific. The areas around the Kerguélen Islands, in the South Pacific between -140°E and -120°E, and around the Fox Islands (North west Pacific, islands east of the tip of

Alaska), in the extreme North Pacific, remain areas of degradation in RMS of the differences in modulus and current direction for the new solution compared with the previous one. On the other hand, the Kuroshio (Japan), the North Atlantic and the Labrador Sea, as well as the Indian Ocean are areas where the RMS of the differences for the new CNES-CLS22 solution has improved compared to the CNES-CLS18 solution.

Globally, the two solutions, CNES-CLS22 and CNES-CLS18, were close considering these drifter comparisons, as shown below (Table 10), where we summarize the drifter comparison statistics over the whole area up to 70°N. Both solutions have a difference RMS between 10.5 and 11 cm/s for U and V components and the current modulus, and a difference RMS of around 34° in direction. The comparison of the two solutions showed that the new one was the best, but by less than 1%. It should also be noted that the comparison between the first guess and the CNES-CLS22 MDT showed an improvement in U and V of around 5% in the RMS of differences, an improvement of around 6% in the current amplitude and 2% in direction, as expected.

Table 10: RMS of the differences for the zonal component U, the meridional component V, the current modulus and the direction of the current between the independent drifting buoy velocities and the altimeter geostrophic velocities obtained using MDT solutions CNES-CLS22, CNES-CLS18 and the CNES-CLS22 first guess. The last two lines show the reduction/increase in RMS of the differences between two solutions: CNES-CLS22 versus CNES-CLS18 and CNES-CLS22 versus CNES-CLS22 first guess, in percent (a negative percentage is a decrease in the RMS of the differences). These statistics were generated using all available drifters up to 70°N.

RMS (differences)	U [cm/s]	V [cm/s]	M [cm/s]	D [°]
CNES-CLS22 MDT	10.57	10.62	10.79	34.24
CNES-CLS18 MDT	10.59	10.63	10.83	34.26
CNES-CLS22 MDT first guess	11.11	11.17	11.47	35.05
% CNES-CLS22 vs CNES-CLS18 MDT	-0.2%	-0.1%	-0.4%	-0.1%
% CNES-CLS22 vs CNES-CLS22 first guess MDT	-4.9%	-4.9%	-5.9%	-2.3%

Validation in the Arctic area

Given the small number of independent drifters in the Arctic, this area was treated separately.

Table 11 summarizes the RMS of the differences between the geostrophic currents derived from drifters (degraded treatment) and the geostrophic currents derived from ADTs calculated with the CNES-CLS22, CNES-CLS18 and CNES-CLS22 first guess solutions. The last two lines showed, as a percentage, the reduction in RMS of the differences between CNES-CLS22 and CNES-CLS18 MDT and its first guess. In the zonal and meridional components of the current, the new solution reduced the RMS of the differences compared with the drifters by 19% and 12% respectively. This improvement translated into a clear improvement in direction of 9%, which reduced the RMS of the differences from around 56° to 50°. However, the RMS of the differences in the current amplitude were only very slightly better for CNES-CLS18. Note that the coverage was not exactly the same, and there were more points taken into account for CNES-CLS22 than for CNES-CLS18.

Table 11:RMS of the differences for the zonal component U, the meridional component V, the current modulus and the direction of the current, between the independent drifting buoy velocities and the altimeter geostrophic velocities obtained using MDT solutions CNES-CLS22, CNES-CLS18 and the CNES-CLS22 first guess. The last two lines show the reduction/increase in RMS of the differences between two solutions: CNES-CLS22 versus CNES-CLS18 and CNES-CLS22 versus CNES-CLS22 first guess, in percent (a negative percentage is a decrease in the RMS of the differences). These statistics are based on the Arctic zone north of 80°N. As the CNES-CLS18 MDT has a smaller coverage, the number of points used is slightly different for this solution than for the others.

RMS (differences)	U [cm/s]	V [cm/s]	Modulus [cm/s]	Direction [°]
CNES-CLS22 MDT	7.95	8.12	9.61	50.43
CNES-CLS18 MDT	9.86	9.23	9.58	55.88
CNES-CLS22 MDT first guess	7.97	8.26	10.02	51.77
% CNES-CLS22 vs CNES-CLS18 MDT	-19.4%	-12.0%	0.4%	-9.8%
% CNES-CLS22 vs CNES-CLS22 first guess MDT	-0.2%	-1.7%	-4.1%	-2.6%

IV.2 Regional Black Sea MDT CMEMS_2020_BLK

The validation consisted in comparisons with independent in-situ measurements from drifters, as detailed in the next sections. Validation results for the MDT_ CMEMS_2020_BLK are fully described in (Jousset et al. 2023).
The regional MDT was also compared to others existing MDTs:

- **MHI MDT**: the observed MDT from Kubryakov and Stanichny (2011), realized from altimetry and in-situ data following a synthetic method close to the one used in the step 2 here.
- **CMEMS MFC MDT**: it is the average of the SSH (Sea Surface Height) over the 1993-2012 reference period derived from the regional CMEMS MFC model BLKSEA_MULTIYEAR_PHY_007_004.

IV.2.1 MDT overview

As shown in Figure 10, the new CMEMS_2020_BLK MDT defines a cyclonic circulation with higher water levels at the periphery than in the centre. The minimum of the gyre is central. This gyre is elongated from west to east. MDT is fairly smooth and does not represent mesoscale structures.

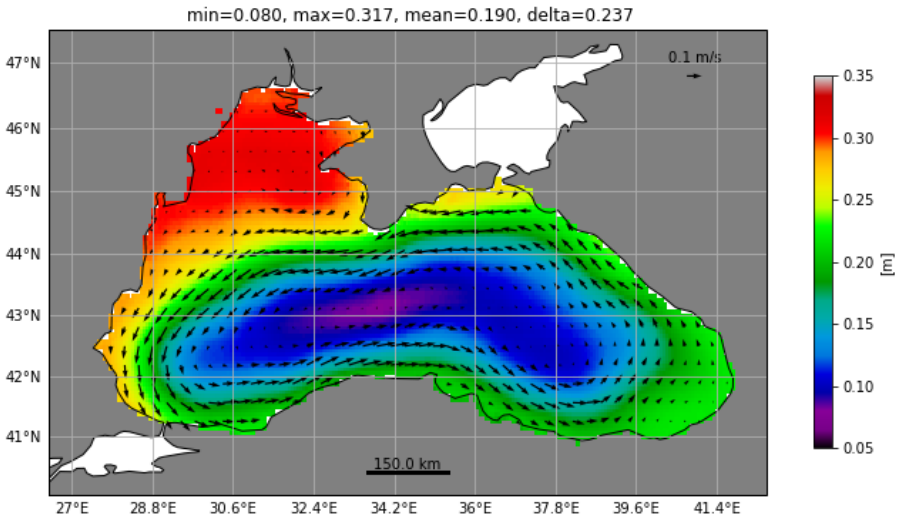

Figure 10: The new MDT_CMEMS_2020_BLK and associated geostrophic currents.Region: Black Sea.

IV.2.2 Mean Dynamic Topography height

The CMEMS_2020_BLK MDT is compared to other MDTs: MHI and CMEMS_Black Sea MFC (Model Forecasting Center) MDTs. Both MDTs are shown in the Figure 11 and are centered around 0 cm instead of 19 cm, as it is the case of the MDT_CMEMS_2020_BLK (Table 12). There is frequent bias between the observed MDT and the MDT from the numerical model. This is also the case for global MDT, see the CMEMS Variable infomation[4]. This is explained because the reference surfaces are not the same. In a numerical model, the reference is given by the bathymetric model, there is no geoid information and geoid height equals 0. For MDT_CMEMS_2020_BLK the reference surface is the geoid height by construction and by definition. Kubryakov and Stanichny (2011) do not used a geoid model for the MHI MDT computation and reference their MDT using a synthetic estimate from the drifters' velocities. They thus integrated the geostrophic velocities estimated from drifters to deduced their synthetic MDT: by definition, their synthetic MDT is defined up to a constant.

[4] https://help.marine.copernicus.eu/en/articles/6025269-what-are-the-differences-between-the-ssh-and-sla-variables

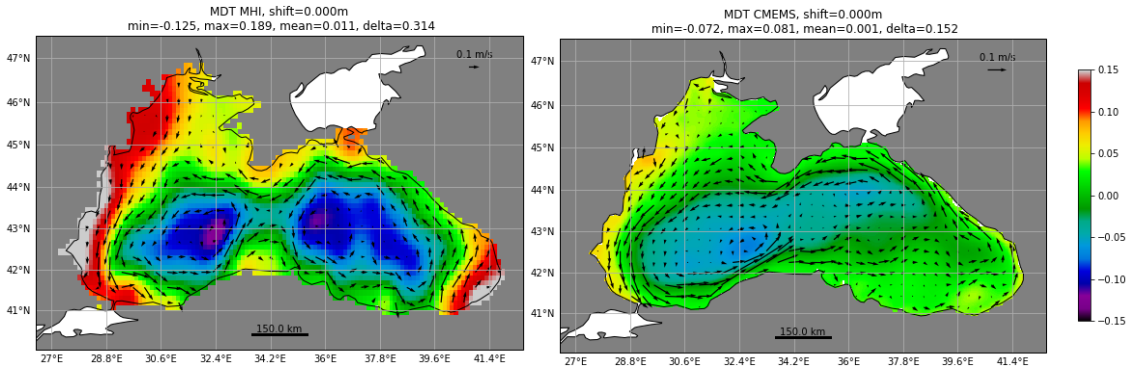

Figure 11: Kubryakov and Stanichny (2011) MHI MDT rereferenced over 1993-2012 period (left), and CMEMS SSH average over 1993-2012 (right). Region: Black Sea.

The MHI MDT calculated over the reference period 1993-1999 was re-referenced over the 1993-2012 period. This MDT defined a cyclonic circulation with two separate cyclonic systems (one in the east and one in the west).

In comparison, the CMEMS MFC MDT is flatter (see delta in the Table 12). The CMEMS MFC MDT represents two gyres, one in the West, with a minimum almost in the center of the basin, and a smaller East gyre (which consists of several cores), located further North. The cyclonic current that circles around the basin follows the coast in the northeast, then follows the bathymetry slope in the northwest, follows again the coast at the west and southwest, but detaches the coast in the southeast.

Table 12 Mean and delta (maximum-minimum) of MDT_CMEMS_2020_BLK, MHI MDT and CMEMS MDT.

Name	Mean [cm]	Delta max-min [cm]
MDT_CMEMS_2020_BLK	19.0	23.7
mdt MHI	1.1	31.4
mdt CMEMS	0.1	15.2

We also compared the MDT_CMEMS_2020_BLK to the dynamic height available in the area estimated from T/S profiles and referred to 500 m. These data were not used in the objective analysis (estimation of the MDT). Figure 12 shows the anomaly of the dynamic heights (with respect to the average over the whole Black Sea). These anomalies were averaged in 0.25° x 0.25° cells. The MDT_CMEMS_2020_BLK was consistent with these data, showing the lowest RMSD (Table 13).

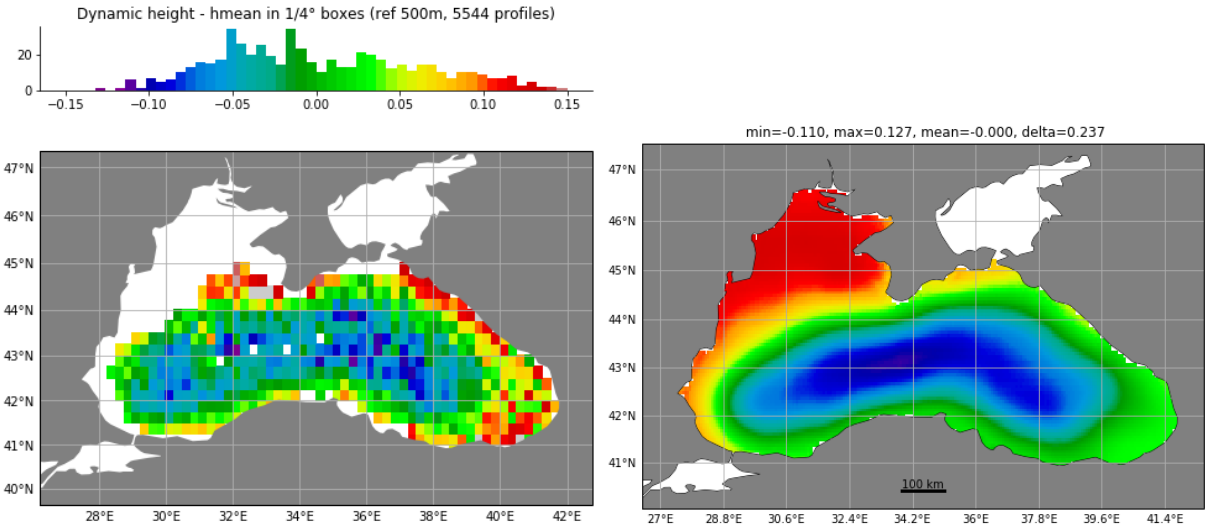

Figure 12: Height anomalies from the T/S profile data (on the left) and from the MDT (MDT-MDT averaged over the whole area) on the right. Region: Black Sea.

Figure 13 represents the RMSD of the different MDTs minus the dynamic heights, measured in 1° x 1° cells. Globally, the new MDT wa better than the two others in the centre of the basin. In the West, it can be noted that the CMEMS MFC MDT performed better in areas close to the coast, the MDT MHI having too high values in this western area.

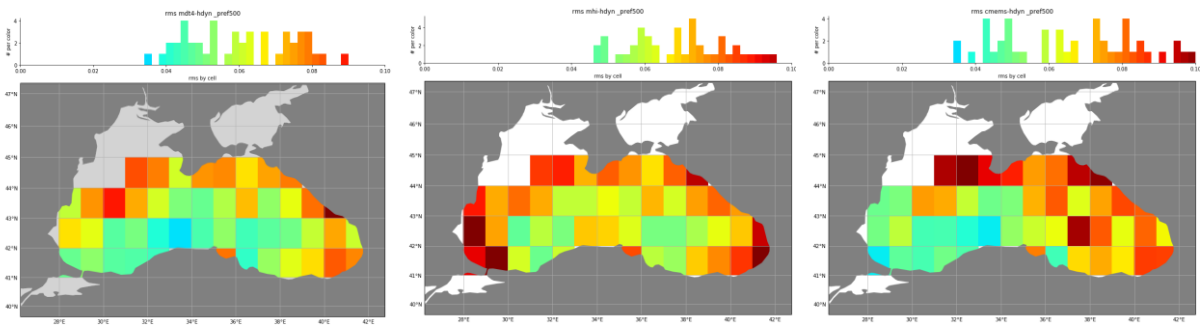

Figure 13: RMS of the differences between the dynamic heights averaged in 1° x 1° cells and the MDT_CMEMS_2020_BLK (left), the MDT MHI (middle) and the MDT CMEMS MFC (right). Minimum 50 observations per cell. Region: Black Sea.

IV.2.3 Mean Geostrophic Currents

From the total set of the drifters dataset, 20% was randomly selected and kept for using in the validation as independent data. Figure 14 shows the 17 drifters kept for the validation of the MDT. There were no drifters in the Northwest part of the Black Sea above the plateau. In addition, there were few observations in the North and the center of the eastern part.

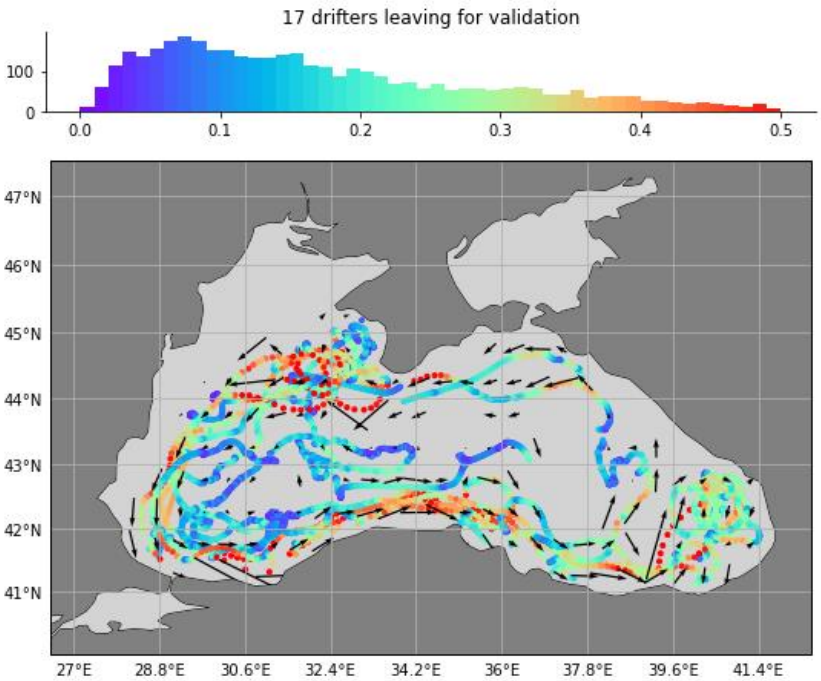

Figure 14: Speed (m/s) of the 17 drifters used for validation. Arrows represent mean velocities of drifters in 2° x 2° cells. Region: Black Sea.

We compare the three MDTs (MDT_CMEMS_2020_BLK, MHI MDT and CMEMS MFC MDT) with the drifters as follows:

- we filtered the drifter trajectories and removed the wind-driven component from the drifter velocities to have an estimate of the geostrophic current (as it was done to compute synthetic velocities in Table 13).
- we added SLA to the three MDTs and derivated the associated geostrophic current.

Table 13 shows RMS of the difference (RMSD) and correlation coefficients between drifters velocities and geostrophics currents associated to MDTs. For both velocities components, the MDT_CMEMS_2020_BLK showed better agreement with the independent drifters than the other MDTs:

- correlation was higher: 0.76 for U component versus 0.73 and 0.70 for other MDTs; correlation of 0.59 for V component versus 0.54 and 0.57 for other MDTs
- RMSD was smaller: 11.50 cm/s for U component versus 12.02 and 12.59 for the other MDTs; 11.20 cm/s for V component versus 12.26 and 11.47 for the other MDTs.

Table 13: RMS of the difference and Pearson correlation coefficients between drifters velocities (zonal component U and meridional component V) and geostrophic currents associated to MDTs over all the area. "Heights RMSD" shows the RMSD between the MDT and the synthetic heights (estimated from the T/S profiles) over the whole area.

Name	**U RMSD [cm/s]**	**U correlation coefficient**	**V RMSD [cm/s]**	**V correlation coefficient**	**Heights RMSD [cm]**
MDT_CMEMS_2020_BLK	11.50	0.76	11.20	0.59	5.92
MDT MHI	12.09	0.73	12.26	0.54	6.98
MDT CMEMS	12.59	0 .70	11.47	0.57	6.79

The Figure 15 represents the RMSD (Root Mean Squared of the difference) in cells of 1° x 1° between the drifting currents and the geostrophic currents estimated from the different MDTs. On the left are the comparisons on the zonal component and on the right the meridional component. From top to down the figure displays the new MDT_CMEMS_2020_BLK, the MDT MHI MDT and the CMEMS MFC MDT.

Regarding the zonal component, the RMSD were globally lower for the MDT_CMEMS_2020_BLK. In particular, MDT_CMEMS_2020_BLK showed better performances than MHI MDT in the center of the basin where the circulations are different. It can also be noted that the RMSD is higher for the CMEMS MFC MDT in the South East, where the cyclonic current breaks away from the coast.

For the meridional component, the RMSD was globally lower for the new MDT_CMEMS_2020_BLK, especially in the centre and compared to MHI MDT, where the system of two very distinct gyres seemed worse compared to the drifters' speeds. In contrast, the MDT MHI performed better in some cells. For example in the far west, where the stronger currents of MHI MDT gave better results.

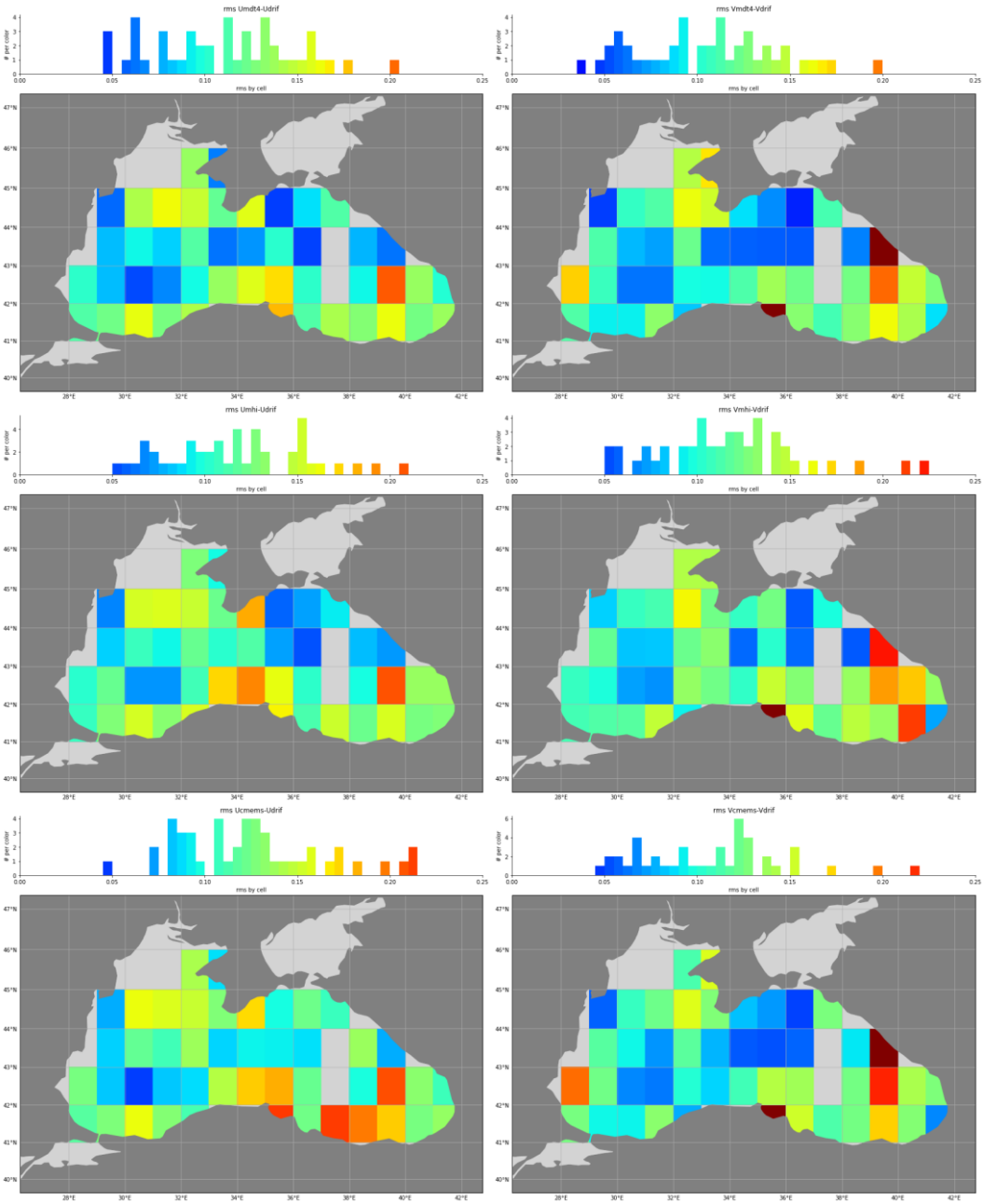

Figure 15: RMSD between drifters (wind corrected) and the geostrophic currents ($U_{mdt} + U_{sla}$). Zonal (left), and meridional (right) components of the currents. New MDT_CMEMS_2020_BLK (top), MHI MDT (middle) and CMEMS MFC MDT (bottom). RMSD in 1° x 1° cells. The scale is the same for all the elements in the figure and ranges from 0 to 0.25 m/s. Region: Black Sea.

IV.3 Regional Mediterranean Sea MDT CMEMS_2020_MED

The validation consisted in the comparison with independent in-situ measurements from drifters and from profiles T/S, as detailed in the next sections. Validation results for the MDT_ CMEMS_2020_MED are fully described in Jousset et al. (2023). The regional MDT was also compared to other existing MDTs:

- **SMDT 2014**: the synthetic MDT from Rio et al. (2014b), derived from the model first guess, altimetry and in-situ data, following a synthetic method similar to that used in step 2 in the present document.
- **CMEMS MFC MDT**: is the average Sea Surface Height (SSH) over the 1993-2012 reference period derived from the regional CMEMS MFC model BLKSEA_MULTIYEAR_PHY_007_004.

As in Rio et al. (2014b), to carry out the validation statistics the Mediterranean Sea was divided into the following zones that are displayed in Figure 16:

- Alboran Sea + Algerian Current
- Baleric Islands Area
- North West Mediterranean basin
- Tyrrhenian + Adriatic Seas
- Ionian Sea
- Levantin Basin

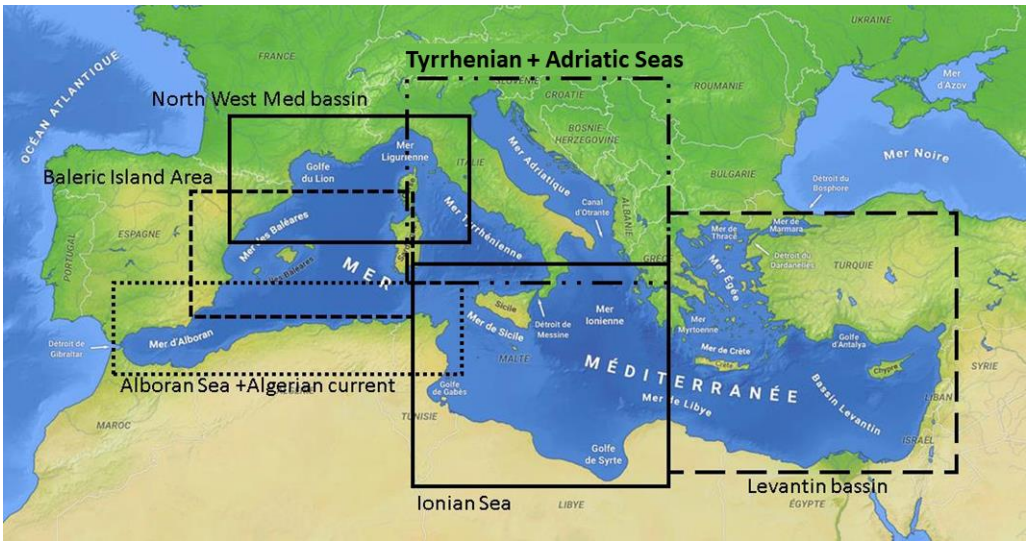

Figure 16: Validation zones (following Rio et al 2014).

IV.3.1 MDT overview

In the top of Figure 17 the new MDT_CMEMS_2020_MED is shown, whereas the associated currents are displayed at the bottom. In this new MDT it is possible to observe recognisable structures such as the Algerian current and the Liguro Provençal current runs north from Palermo along the Italian coast before heading west near Monaco, passing through the Gulf of Lion and reaching the Catalan coast. In the new MDT, the Pelops, Cretan Cyclone, Irapetra and Rhodes eddies showed a remarkable signature in the eastern basin. We can also note the northward coastal current at the east of the Levantine basin.

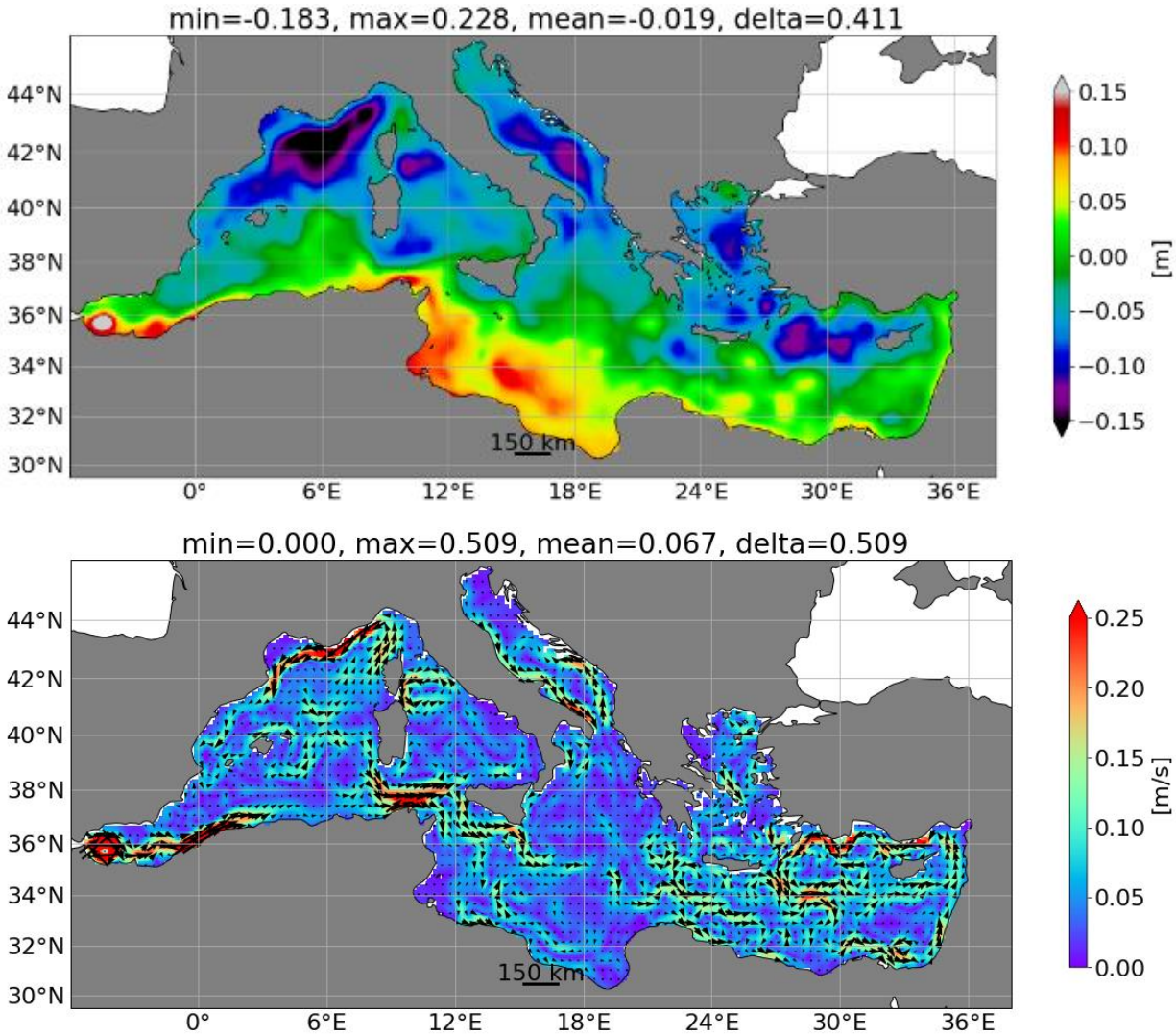

Figure 17: New MDT_CMEMS_2020_MED (top) and the geostrophic circulation associated (bottom). Region: Mediteranean Sea.

IV.3.2 Mean Dynamic Topography height

From the total dynamic heights dataset, a random subset comprising 15% of the data was kept to be used in the validation as independent data. This validation dataset has been randomly collected since 2012 in order to have an independent validation data subset of the new MDT but also of the SMDT2014. The Figure 18 shows the independent data on the dynamic heights kept for the validation of the MDT. The dynamic heights were computed from available hydrological profiles referred to 350 m. This reference depth is the same used in the SMDT2014. The deeper the reference depth, the lesser number of available profiles. However, this lower number of profiles are able to better capture the baroclinic content. For this reason, the choice of the reference depth (350 m) was based on a balance between the number of available profiles and the dynamical content of their calculated dynamic heights.

As a consequence, there were neither data for the coast of Libya and Tunisia, nor at the bottom of the Adriatic Sea, as these areas are on the continental shelf where the depth is less than 350 m.

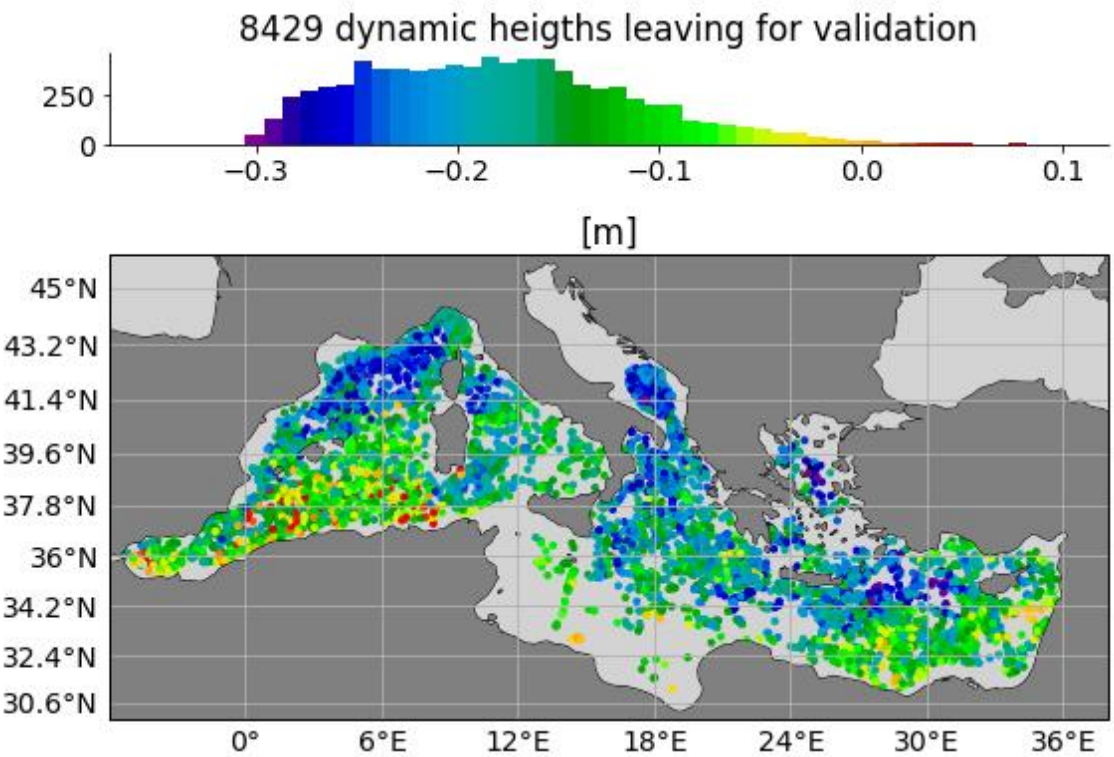

Figure 18: Random subset (15% of the data, collected from 2012 to 2018) from the total dynamic heights dataset used as independent data for validation. The dynamic heights were computed from available hydrological profiles referred to 350 m. Region: Mediterranean Sea.

In Figure 19 we compared the dynamic heights at which SLA was removed (averaged values in 1° x 1° cells) with the MDTs at the east of the Mediterranean Sea. In order to compare these heights, we have also removed the average over the entire Mediterranean (heights centered around zero and MDTs centered around zero). In the Aegean Sea, the new MDT seemed a little better than the SMDT2014 regarding the global height of the sea. However, there were not enough data to validate the

represented structures. Improvements were more marked around Crete, with a more consistent representation that allowed the detection of patterns south of Crete (e.g., Irapetra and Rhodes eddies). On the other hand, as we have already seen with drifters above, the coastal currents were less intense in the new MDT, although we could not really compare heights, because there were not enough data in this coastal area.

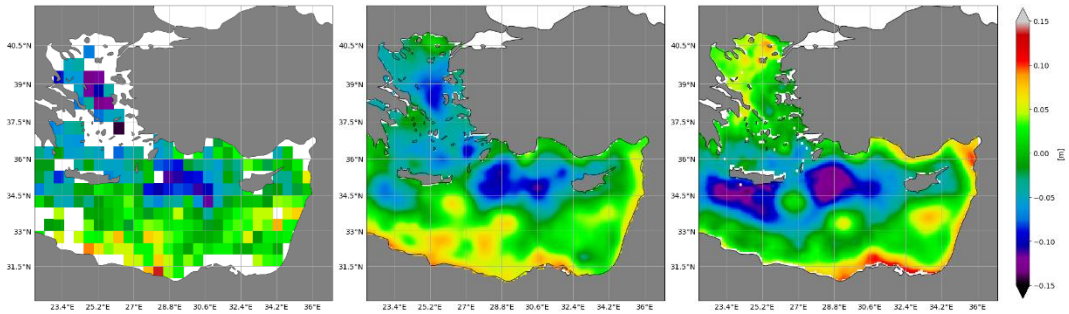

Figure 19: Left: dynamic height data corrected by SLA and averaged by cells of ¼° x ¼°; Middle: solution of the new MDT (mean is recentered on zero); Right: solution of SMDT2014 (mean recentered on zero). Region: East of the Mediterranean Sea.

Figure 20 shows the Pearson correlation coefficients and RMSD scores by geographical zone between MDT height (Absolute Dynamic Topography (ADT = MDT + SLA) compared to dynamic heights deduced from in-situ measurements. Conclusions are the same: in all zones, correlation coefficients are higher for the new MDT than for the SMDT2014 and the RMSD is lower.

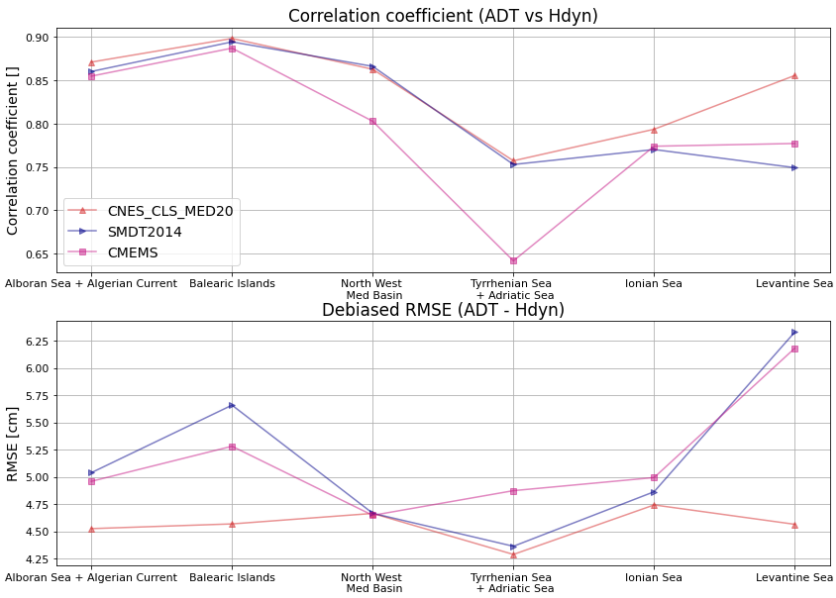

Figure 20: Correlation coefficients by geographical area (top) between dynamic heigths and Absolute Dynamic Topography (ADT = MDT + SLA). RMSD scores (bottom) by area between ADT (MDT + SLA) and dynamic heigths. The results are represented in different colours for the new MDT_CMEMS_2020_MED, the SMDT2014 and the MDT model (average of the SSH of the CMEMS model over the reference period).

Finally,Table 14 gives the comparison statistics with in-situ data (heights and velocities) over the whole Mediterranean Sea for the new MDT_CMEMS_2020_MED, SMDT2014 and MDT CMEMS MFC. The conclusions remain identical to those of the area comparisons.

Table 14: RMS of the difference and Pearson correlation coefficients between drifters velocities (zonal component U and meridional component V) and geostrophic currents associated to MDTs over all the area. In addition, the RMSD and correlation coefficients between the MDT and the synthetic heights are provided.

Name	U RMSD [cm/s]	U correlation coefficient	V RMSD [cm/s]	V correlation coefficient	Heights RMSD [cm]	Heights correlation coefficient
MDT_CMEMS_2020_MED	10.38	0.70	11.47	0.59	5.06	0.85
SMDT2014	10.90	0.68	11.89	0.57	5.33	0.85
CMEMS MFC	11.01	0.66	12.11	0.54	5.29	0.82

IV.3.3 Mean Geostrophic Currents

From the total dataset of drifters, 15% of the data were kept for the validation as independent data. This validation dataset contains the validation set from the drifters of SMDT 2014 (drifters from July to December 2011) and drifters randomly chosen since 2012 in order to have an independent validation set for the new MDT but also of SMDT2014.

The Figure 21 shows the drifters kept for the validation of the MDT. There are no drifters in the East of the Tyrrhenian Sea and in the Aegean Sea, and only a few drifters in Levantin Basin and offshore Libya.

Thereafter, to compare to the MDT, we filtered the data at 36 h to remove the tide and inertial waves, then we remove the wind-driven current (estimated with method and parameters from Poulain et al. (2012) to keep only the geostrophic signal and compare it to the geostrophic currents (U_{mdt} +U_{sla}).

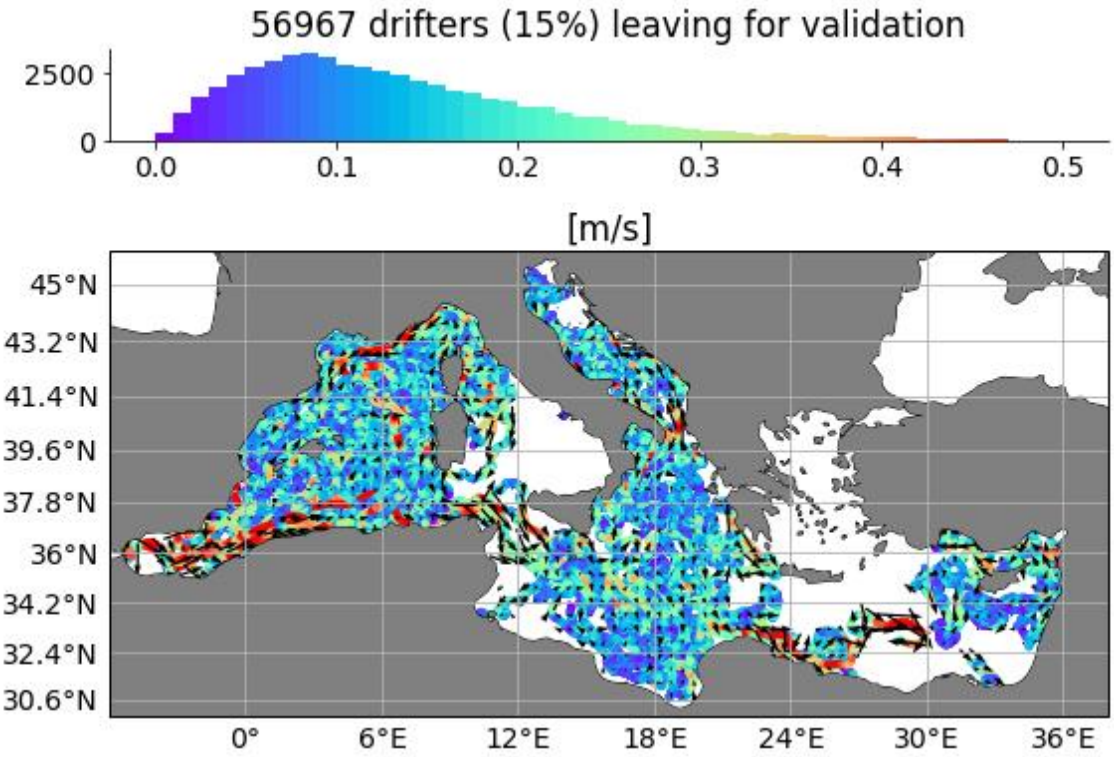

Figure 21: Drifters left for validation. Region: Mediterranean Sea.

Figure 22 shows the average circulations of the validation drifters and the two MDTs in the southwest part of the Mediterranean Sea. At the top is shown the mean speed per 1° x 1° cells of the drifters currents from which the wind-driven current and the geostrophic current derived from altimetry (SLA) have been removed. In the middle (respectively in the bottom) are represented the mean geostrophic velocities associated with the new MDT_CMEMS_2020_MED (respectively SMDT2014). The Algerian current is more intense in SMDT 2014, especially between longitudes 2°E and 8°E. In the MDT_CMEMS_2020_MED, it almost disappears between longitudes 4°E and 8°E, in agreement with drifters. Indeed in this zone, eddies can drive the drifters towards the North (observed on drifters trajectories not shown here).

The difference with the SMDT 2014 is certainly due to the fact that the MDT_CMEMS_2020_MED first guess (MSS - geoid) is smother than the SMDT2014 first guess (from the numerical model), probably less accurate close to the coast and does not represent this very coastal current while it was present in the first guess model of the SMDT2014.

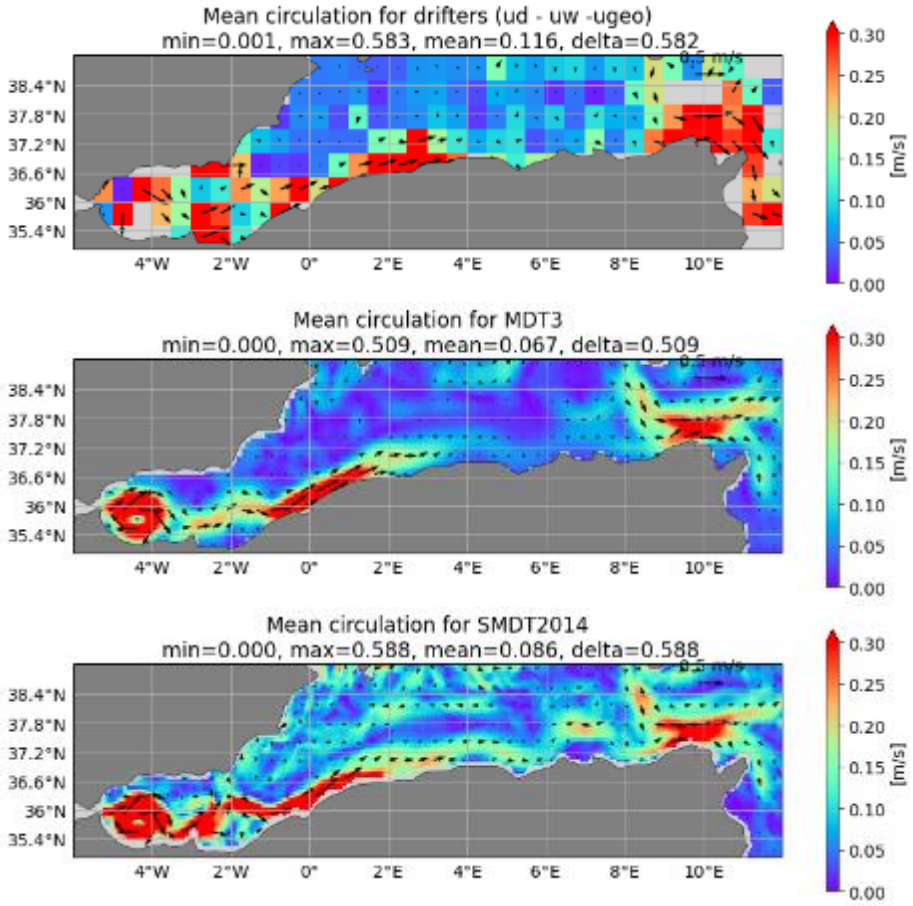

Figure 22: Mean speed in 1°x 1° cells. Top: drifters currents from which the wind-driven current and the geostrophic current derived from altimetry (SLA) have been removed. Middle: average circulation associated with the MDT_CMEMS_2020_MED. Bottom: average circulation associated with the SMDT2014. Region: southwest part of the Mediterranean Sea.

Figure 23 shows the Pearson correlation coefficients and RMSD scores by geographical zone between wind corrected drifters speeds and geostrophic speeds derived from MDT and SLA (zonal component on the left and meridional component on the right). The results for the new MDT_CMEMS_2020_MED, the SMDT2014 and the CMEMS MFC MDT model (average of the SSH of the MED MFC CMEMS model MEDSEA_MULTIYEAR_PHY_006_004 over the reference period) are represented in different colours. In all zones, for both current components the correlation coefficients were higher and the RMSD was lower for the new MDT_CMEMS_2020_MED compared to the SMDT2014. Figure 20 shows similar comparison but for height (specifically the ADT compared to dynamic heights). Conclusions derived from speeds and heights were similar: in all zones, correlation coefficients were higher and the RMSD was lower for the new MDT compared to the SMDT2014.

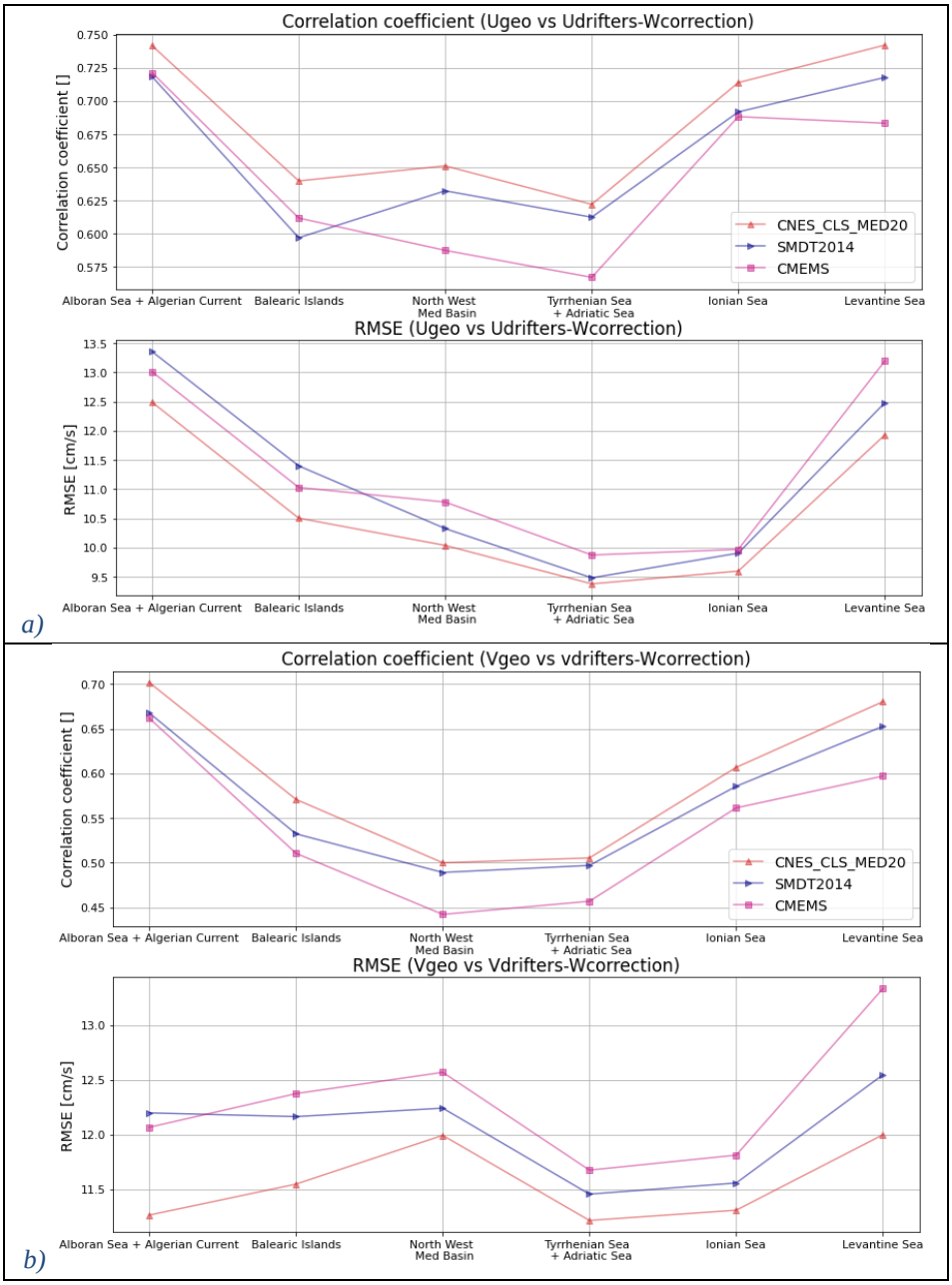

Figure 23: (a) For the zonal component of the current, correlation Pearson coefficients by geographical area between wind corrected drifters speeds and geostrophic speeds associated with Absolute Dynamic Topography (ADT = MDT + SLA) are shown on top and RMSD scores by area between the geostrophic velocities (MDT + SLA) and the wind corrected drifters velocities are shown on bottom. (b) Same plots for th meridional component of the current. In different colors are represented the results for the new MDT_CMEMS_2020_MED, the SMDT2014 and the MDT model (average of the SSH of the CMEMS model over the reference period).

Table 14 above summarises the statistics for comparison with in-situ data on heigths over the whole Mediterranean Sea for the new MDT_CMEMS_2020_MED, SMDT2014 and MDT CMEMS MFC. The conclusions remain identical to those of the area comparisons.

IV.4 Regional European coasts MDT CMEMS_2024_EUR

The validation consisted in a comparison with independent drifters, as detailed in the next sections.
The regional MDT was also compared to others existing MDTs:

- **CNES-CLS22 MDT**: the global MDT (section I.1.1).
- **COMBAT MDT**: regional MDT calculated in the Bay of Biscay with HF radar data (Caballero et al. 2020)
- **IBI system MDT**: is the average over the 2013-2019 period re-referenced on the 1993-2012 reference period (with DUACS) of the Sea Surface Height (SSH) from the regional CMEMS MFC model free run.
- **MERCATOR HYBRIDE MDT (V6)**: the new hybrid MDT is derived from an ensemble method based on MDTs: mean of GLO12V4 SSH over 2013-2021, CNES-CLS22 (section I.1.1) and DTU22.

IV.4.1 MDT overview

The CMEMS_2024_EUR MDT has been calculated on the European coasts, in the Mediterranean Sea and in the Black Sea are the CMEMS_2020_MED (section I.1.2) and CMEMS_2020_BLK (section I.1.2) solutions, respectively. The new CMEMS_2024_EUR solution is shown in Figure 24a and the associated geostrophic currents in Figure 24b.

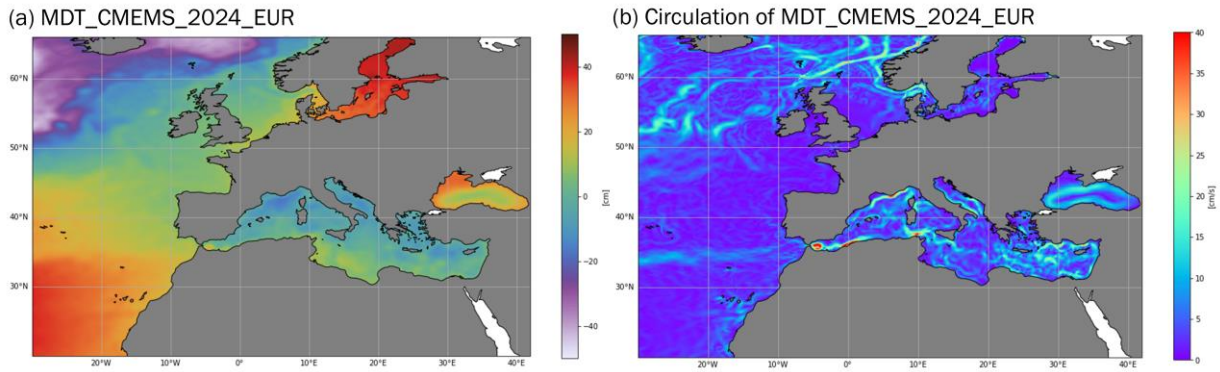

Figure 24: MDT_CMEMS_2024_EUR (a) and geostrophic circulation associated (b). Region: Europe.

IV.4.2 Mean Dynamic Topography height

Figure 25 shows the fields of the different MDTs over the extended Bay of Biscay area, where there is an impact of HF radar data available over the cells comprising the area between 3°W to 1°W and 43.25°N to 44.5°N (area restricted to the south-east of the Bay of Biscay). These were processed as part of the COMBAT project (Caballero et al. 2020) as inputs to the CMEMS_EUR_2024 MDT and the COMBAT MDT (but not to the other MDTs). The CMEMS_EUR_2024 and CNES-CLS22 MDTs were very similar, although the East-West gradient of MDT near the French coast between 43.5°N and 44.5°N was weaker in the CMEMS_EUR_2024 MDT than in the CNES-CLS22 MDT due to the HF radar data.

The COMBAT MDT showed larger MDT gradients along the Spanish and French coasts than the other MDTs presented, except for the IBI system MDT gradient along the French coast. The reason for these gradients is that the first guess worked specifically in this area for this MDT. The gradient along the Spanish coast created a current that seems too wide compared to other MDTs.

In contrast to the MDTs referenced on the geoid, the two MDTs based on the numerical models (Figure 25d MERCATOR HYBRIDE MDT and Figure 25e IBI systeme MDT) were referenced in a disimilar way, thus there is a bias between these different MDTs. In order to facilitate comparisons, the bias (about 40 cm) was added to these two model MDTs in Figure 25. The IBI system MDT seems to have more intense gradients close to the coast but also globally over the whole area than the other MDTs.

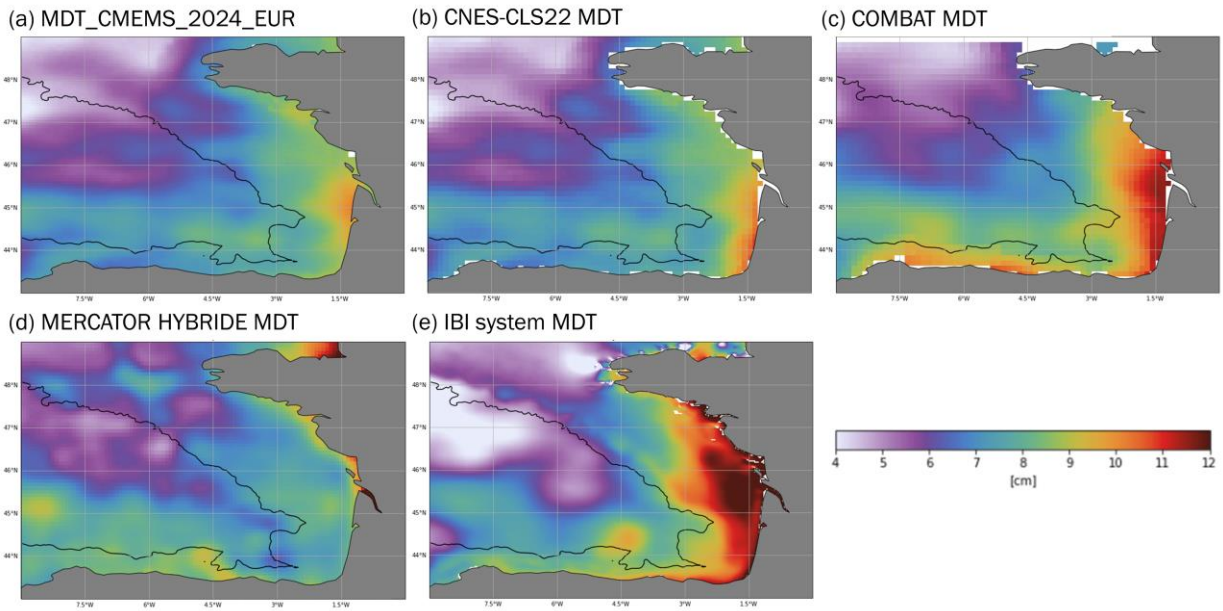

Figure 25: (a) CMEMS_2024_EUR MDT, (b) CNES-CLS22 MDT, (c) COMBAT MDT, (d) MERCATOR HYBRIDE MDT and (e) IBI systeme MDT on extend Bay of Biscay area. The black line indicates the 2000 m isobath. The two MDTs based on numerical models -(d) MERCATOR HYBRIDE MDT and (e) IBI systeme MDT- are not referenced on the geoid as the other MDTs are, thus there is a bias between these different MDTs. In order to facilitate comparisons (using the same scale bar), the bias (about 40 cm) was added to these two model MDTs.

To view the energy content of the MDT, the different MDT solutions were studied using spectral analysis. Figure 26 shows the spectral analysis carried out over the extended Bay of Biscay area, between -9°E and -1.5°E and between 43.7°S and 46.5°S. The three solutions based on geodetic MDT and in-situ observations (CMEMS_2024_EUR, CNES-CLS22 and COMBAT) showed very similar energy content in this zone, between 300 and 50 km. Below 50 km, the energy descent of these three solutions is lower. The COMBAT MDT is the solution which seems to reach a plateau, certainly linked to noise, at around 40 km.

As the CMEMS_2024_EUR MDT is at higher resolution, we can see that this solution reached an energy plateau around 20 km. For the CMEMS_2024_EUR MDT, residual scales below 60 km, below 40 km and below 20 km were observed (figures not shown). The MDT fields and associated geostrophic currents showed coherent physical structures, but at smaller scales the noise becomes visible (e.g., the residual scales below 20 km are noisy).

The two solutions based on or using a numerical model showed more energy between 40 and 300 km, and the slope was more pronounced at smaller spatial scales. Compared to these solutions with a numerical model input, we can see that the CMEMS_2024_EUR MDT lacked of small scales (< 20 km).

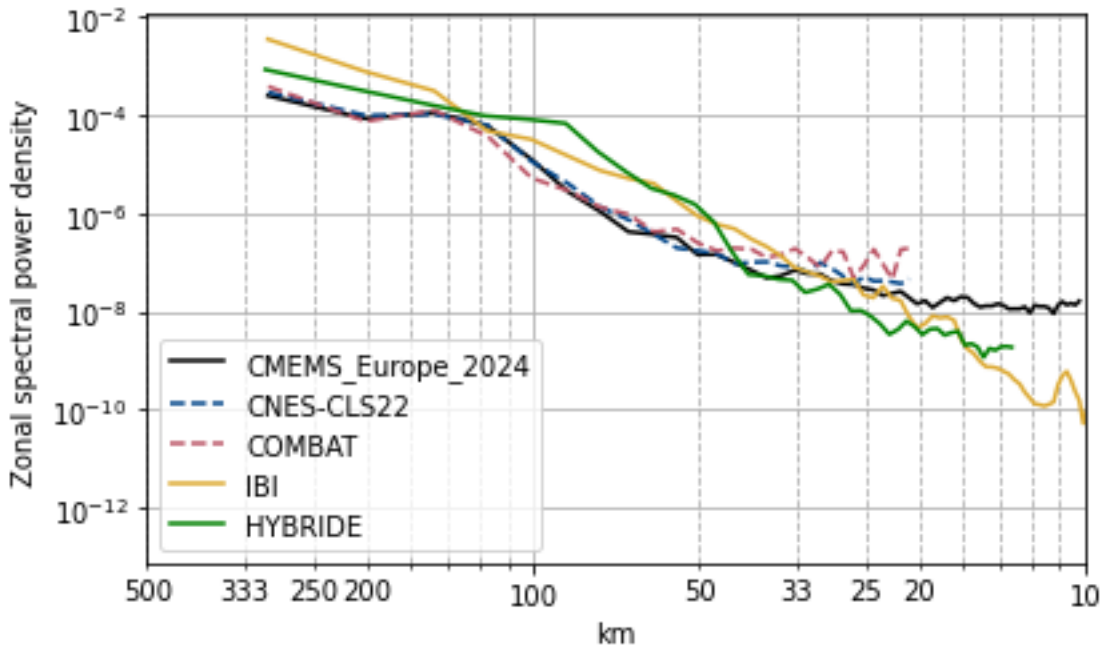

Figure 26: Spectral Power Density obtained for different MDT solutions in the Bay of Biscay area, specifically between -9°E and -1.5°E and between 43.7°S and 46.5°S.

IV.4.3 Mean Geostrophic Currents

Figure 27 shows the fields of different geostrophic circulation associated with different MDTs over the extended Bay of Biscay area, where there is an impact of HF radar data available over the cell 3°W to 1°W and 43.25°N to 44.5°N (area restricted to the south-east of the Bay of Biscay) as inputs to the CMEMS_EUR_2024 MDT and the COMBAT MDT, but not the other MDTs.

The eastward current along the Spanish coast according to the IBI system MDT was intense but very close to the coast. For the MERCATOR HYBRIDE MDT, this current was not marked to the west, but it was intense to the east. For the COMBAT MDT, this current was well marked to the east, but fairly wide. For the CMEMS_EUR_2024 MDT, the current was visible in the eastern part, but interrupted in the western part (with a current branch to the west due to drifter data).

In the area observed by HF radars, the CNESCLS22 MDT had a relatively strong northerly current and was very wide, which was not the case in the radar data. The CMEMS_EUR_2024 MDT offered a solution closer to the radars. The MDT COMBAT was very similar to the radar data in the area, but it created a discontinuity at about 45°N.

Along the continental slope, the IBI system MDT has a north-westerly current, which is also present but more sinuous in the MERCATOR HYBRIDE MDT. In the CNES-CLS22 and in the CMEMS_EUR_2024 MDT, this current is slightly visible in the south-east zone. However, further to the north-west, this current is not represented at all, and in fact flows in the opposite direction. In MDT COMBAT, this current is not present. As a reminder, these structures have currents of just a few cm/s, making them difficult to represent. The structures are different in the CNES-CLS22 MDT and the CMEMS MDT.

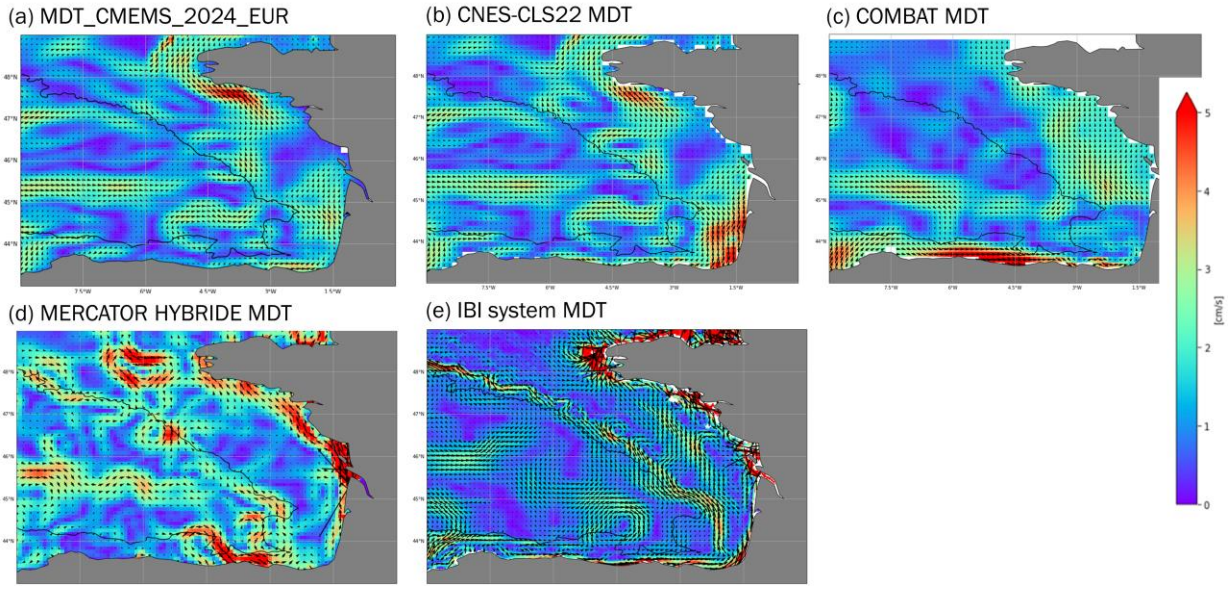

Figure 27: Geostrophic circulation associated with (a) CMEMS_2024_EUR MDT, (b) CNES-CLS22 MDT, (c) COMBAT MDT, (d) MERCATOR HYBRIDE MDT and (e) IBI systeme MDT on extend Bay of Biscay area. The black line represents the 2000 m isobath.

Globally, 10% of the AOML drifters dataset are randomly selected and kept for use in validation as independent data. Here we use these independent drifters, which are not used as inputs to the CMEMS_2024_EUR MDT. In the section on the global MDT, Figure 7 (a) shows the number of measurement points in ¼° x ¼° cells of independent drifters kept for validation and Figure 7 (b) shows the modulus of mean current in ¼° x ¼° cells. It should be noted that there were no validation drifters on the North Sea continental shelf and in the Baltic Sea (two areas that are rarely observed). To allow comparisons with the MDTs, we removed the wind-driven current (Ekman current and wind slippage for undrogued drifters) from the total drifters current. Then the data were filtered at inertia frequency to remove the tide and inertial waves. The objective was to keep only the geostrophic signal. Absolute dynamic topography values were calculated by adding the CMEMS gridded SLA to the CMEMS_2024_EUR MDT and to CNES-CLS22 MDT. Associated geostrophic current were then derived and interpolated along the drifter trajectories.

Root Mean Square differences (RMSD) between the obtained geostrophic velocities and the drifter derived geostrophic velocities were then calculated. The RMSD to the independent velocity dataset was then calculated as a function of the distance to the coast (Figure 28). Results for the CNES-CLS22 are shown in blue and for the CMEMS_2024_EUR in green for modulus -Figure 28 (a)- and for direction -Figure 28 (b)- of the current respectively. CMEMS_2024_EUR MDT results were very similar or even slightly better to CNES-CLS22 MDT in terms of current amplitude and direction. In addition, we can note that RMSD in current amplitude and direction increased sharply within 25 km of the coast.

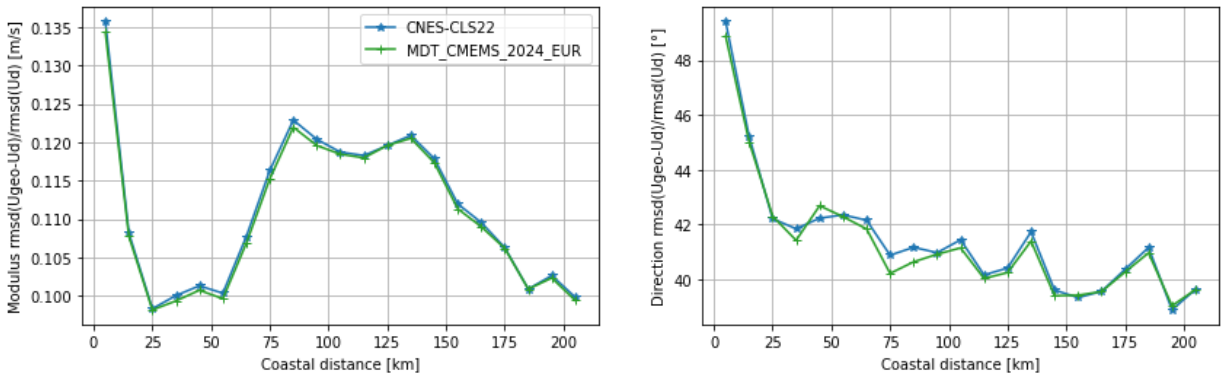

Figure 28: Modulus (left) and direction (right) RMS differences as a function of coastal distance between the independent drifting buoy velocities and the altimeter geostrophic velocities obtained using different MDT solutions. Blue: new CNES-CLS22 MDT; Green: CMEMS_2024_EUR MDT.

Comparisons of RMS of the differences in current modulus (Figure 29 (a)) and in direction (Figure 29b) of the independent drifting buoy velocities and the altimeter geostrophic velocities obtained using MDT solutions (CMEMS_2024_EUR and CNES-CLS22) are shown as a percentage. In blue-red scale, we show the RMSD changes (reductions or increases) when using CMEMS_2024_EUR compared with CNES-CLS22.

In terms of current amplitude -Figure 29 (a)-, MDT CMEMS_2024_EUR is better than CNES-CLS22 in the northern part of the zone; in the western part of the zone there is no clear improvement. The histogram in top of the figure shows that there were more cells where CMEMS_2024_EUR was better than CNES-CLS22. In terms of current direction (Figure 29 (b)), CMEMS_2024_EUR showed improvements in the north, while CNES-CLS22 performed better in the west area of the Iberian Peninsula.

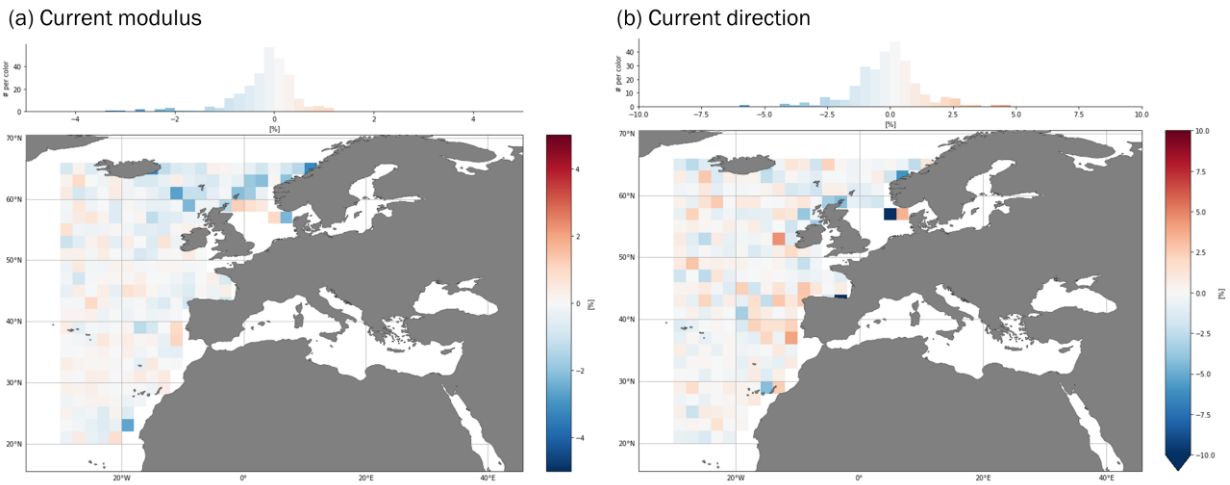

Figure 29: Comparison of the RMS of the differences (a) in current modulus and (b) in direction, of the independent drifting buoy velocities and the altimeter geostrophic velocities obtained using MDT solutions CMEMS_2024_EUR and CNES-CLS22, in percent: $\%RMSD = RMS(U_{ADT24}) - RMS(U_{ADT22} - U_d)/RMS(U_{ADT22} - U_d)$. In blue , the bias is reduced (whereas in red, the bias is increased) using CMEMS_2024_EUR compared with CNES-CLS22. All these statistics are calculated in 2° x 2° cells, and only cells with at least two different difters and at least 50 measurement points, are kept. Region: Europe.

The two solutions, CMEMS_2024_EUR and CNES-CLS22, showed similar results according to these drifter comparisons, as shown in the following Table 15, which summarises the drifter comparison statistics over the whole area. Both solutions have a difference RMS of 8.9 cm/s for U and V components, of 9.4 cm/s for current modulus, and a difference RMS of around 37° in direction.

Table 15: RMS of the differences for the zonal U component, the meridional V component, the current modulus and the direction of the current, of the independent drifting buoy velocities and the altimeter geostrophic velocities obtained using MDT solutions: CNES-CLS22, CNES-CLS18 and the CMEMS_2024_EUR MDT.

RMS (differences)	U [cm/s]	V [cm/s]	M [cm/s]	D [°]
CNES-CLS22 MDT	8.905	8.891	9.456	37.001
CMEMS_Europe_2024 MDT	8.887	8.887	9.430	36.973

V

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

Date	Change/Event description	System version
Sept-2021	New products on Mediterranean and Black Seas: SEALEVEL_MED_PHY_MDT_L4_STATIC_008_066 (see Table 12) SEALEVEL_BLK_PHY_MDT_L4_STATIC_008_067 (see Table 13)	1.0
Jul-2023	Updated of the global MDT: SEALEVEL_GLO_PHY_MDT_008_063 product Changes summarized in Table 11.	1.0
May-2024	New product: SEALEVEL_EUR_PHY_MDT_L4_STATIC_008_070 (see Table 14)	1.0

VI QUALITY CHANGES SINCE PREVIOUS VERSION

Global MDT_HYBRID_CNES_CLS22_CMEMS2020

The changes in methodology and data used to calculate the new global MDT are summarized in

Table 11. The main improvement of the new CNES-CLS22 MDT over the previous CNES-CLS18 MDT is in the Arctic, with better coverage, no artifacts and a more realistic solution.

When comparing the CNES-CLS22 MDT with independent height observations from T/S profiles, we observe a global improvement in the CNES-CLS22 MDT compared with CNES-CLS18, but this is not true in all regions. South of the Atlantic and Indian Oceans (up to Australia), the variance of differences is reduced for CNES-CLS22 by more than 10% in many cells. Areas where the variance of differences with observations is increased are concentrated in the north-western Atlantic Ocean, particularly south of Greenland, in the far north of the Pacific Ocean (in the Gulf of Alaska, and close to Russia) and in the southern Pacific, where an increase in the variance of differences of 10% is observed.

Globally, the two solutions, CNES-CLS22 and CNES-CLS18, were close considering independent drifter comparisons. The comparison of the two solutions showed that CNES-CLS22 was the best, but by less than 1%.

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